Curriculum Vitae of Fahad Saeed, Ph.D.

Full Professor, Knight Foundation School of Computing & Information Sciences (KFSCIS),
Graduate Program Director, KFSCIS
Director, Precision Computational Health and Biomedical Data Science Lab
Full Professor (courtesy), Biomolecular Sciences Institute (BSI),
Full Professor (courtesy), Herbert Wertheim School of Medicine,
Florida International University (FIU), Miami, FL, USA

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Contents

1. Current Fields of Interests and Research Activities

Executive Summary of Research, Teaching and Service activities:
Fahad Saeed is a Full Professor and Graduate Program Director (GPD) in the Knight Foundation School of Computing and Information Sciences (KFSCIS) at Florida International University (FIU), Miami FL. He is also the director of Precision Computational Health and Biomedical Data Science Lab (https://pcdslab.github.io). Prior to joining FIU, he was an Assistant Professor (2014-2018) in the Department of Electrical & Computer Engineering and Department of Computer Science at Western Michigan University (WMU), Kalamazoo Michigan. He was tenured and promoted to the rank of Associate Professor at WMU in July 2018. Dr. Saeed was a Research Fellow in the National Institutes of Health (NIH) from 2011 to 2014 and a postdoctoral fellow from 2010-2011. He received his Ph.D. in the Department of Electrical and Computer Engineering, University of Illinois at Chicago (UIC) in 2010. He has served as visiting scientist in world-renowned institutions such as ETH Zurich, Swiss Institute of Bioinformatics (SIB) and National Institutes of Health (NIH). He has published 100+ peer-reviewed research papers in leading proceedings, and journals, 3 book chapters, editor for 4 conference proceedings, 3 special issue journals, and 6 US Patents. He also co-authored a book in Fall 2022 (Springer, 140 pages).

Dr. Saeed's research interests are at the intersection of machine-learning, high performance computing and real-world applications, especially in computational biology. Currently, Saeed Lab develops machine-learning models, combined with high-performance computing, and data science approaches, to study functional genomics and organization of the human brain. His work focuses on understanding the stochastic difference between identified peptides from high-throughput mass spectrometry data for applications related to human health, and environment. In addition, his work focuses on understanding brain function in the context of prediction, diagnosis and characterization of disorder specific biomarkers for epilepsy, ADHD, Autism, and Alzheimer's. More info can be found at https://pcdslab.github.io and https://prof-s.github.io/.

To accomplish these research activities, Prof. Saeed has acquired some of the most competitive and highly prestigious federal grants in the US including NSF CRII, NSF CAREER[1] award, NIH R01[2], and NIH MIRA award. This is a testament and an acknowledgement by his peers and federal agencies about highly competitive and cutting-edge leadership in the interdisciplinary field of machine-learning and computational biology. On average, he continues to raise **US$ 700,000 per year** as a PI since he acquired the rank of Associate Professor in 2018. Dr. Saeed has been awarded over **US$ 6.85 million** in external research funds - with more than **US$ 5.45 million** as a PI. External fund raising (**US$ 10 Million** + **US$ 6 Million**), Intramural funds (**US$ 895,107**) and computing allocations are not included in this amount.

Prof. Saeed is also an active member of the research community. He has served as program chair, workshop chair, and program committee member for various IEEE/ACM academic conferences. Further, he serves on the editorial board of Springer Journal of Network Modeling Analysis in Health Informatics and Bioinformatics, Associate Editor at Journal of the American Society of Nephrology (JASN), and more recently was elected as (academic) Editor for PLoS Mental Health. Dr. Saeed is a senior member of ACM and a Senior Member of IEEE. His honors include ThinkSwiss Fellowship (2007, 2008), NIH Postdoctoral Fellowship Award (2010), Fellows Award for Research Excellence (FARE) at NIH (2012), NSF CRII Award (2015), Outstanding New Researcher Award at WMU (2016), NSF CAREER Award (2017), WMU Distinguished Research and Creative Scholarship Award (2018), FIU KFSCIS Excellence in Applied Research Award (2020), and FIU Top Scholar Award (2022) for Research and Creative Activities.

[1] NSF Faculty Early Career Development (CAREER) Award is considered the most prestigious award in support of early- career faculty who have the potential to serve as academic role models.

[2] NIH R01 grant is the most sought after, and highly competitive grant mechanism at NIH, in the US and worldwide.

2. Education and Professional Preparation

Degree	Institution	Field	Dates
Research Fellow	National Institutes of Health (NIH), Bethesda MD Advisor: Mark Knepper, MD/PhD	Computational Systems Biology and Bioinformatics	08/2011 – 01/2014
Post-Doctoral Fellow	National Institutes of Health (NIH), Bethesda MD Advisor: Mark Knepper, MD/PhD	Computational Systems Biology and Bioinformatics	08/2010 – 08/2011
PhD	University of Illinois at Chicago (UIC), Chicago IL Advisor: Ashfaq Khokhar, PhD	Electrical and Computer Engineering	08/2006 – 08/2010
BSc Engineering	University of Engineering & Technology (UET) Lahore Advisor: Shahid H. Bokhari, PhD	Electrical Engineering	01/2002 – 01/2006

3. Work Experience

3.1. Full-Time Academic Experience

Institution	Role	Department/Field	Dates (Month & Year)
Florida International University (FIU), Miami FL USA	Full Professor	Knight Foundation School of Computing & Information Sciences (KFSCIS)	08/2025 - to date
	Graduate Program Director	KFSCIS	05/2025 - to date
	Associate Professor (courtesy)	Biomolecular Sciences Institute (BSI)	03/2021 – to date
	Associate Professor (courtesy)	Department of Human and Molecular Genetics, Herbert Wertheim College of Medicine	06/2021 – to date
	Director	Precision Computational Health and Biomedical Data Science Lab (Saeed Lab)	08/2018 - to date
	Associate Professor (Tenured)	KFSCIS	08/2018 – to date
Western Michigan University (WMU), Kalamazoo MI USA	Associate Professor (Tenured)	Department of Computer Science	08/2018 – 08/2018
	Center Co-Director	WMU High Performance Computing	10/2014 – 08/2018
	Assistant Professor	Department of Computer Science (50%), Department of Electrical and Computer Engineering (50%)	01/2014 – 08/2018
National Institutes of Health (NIH), Bethesda MD USA	Research Fellow	Computational Systems Biology Center, NHLBI	06/2011 – 01/2014
	Postdoctoral Fellow	Epithelial System Biology Lab, NHLBI	08/2010 – 06/2011
University of Illinois at Chicago (UIC), Chicago IL	Graduate Research Assistant	High Performance Computing and Bioinformatics at ECE Department	08/06 - 08/2010
Space and Upper Atmosphere Research Commission	Assistant Manager (Research)	Scientific Software Development	12/2005 - 08/2006

(SUPARCO), Lahore Pakistan			
University of Engineering & Technology (UET), Lahore Pakistan	Undergraduate Research Assistant	Electrical Engineering Department	01/2005 - 12-2005

3.2. Part Time Academic Experience

Institution	Role	Field	Dates (Month &Year)
National Institutes of Health (NIH), Bethesda MD USA	Visiting Scientist	Computational Biology/Bioinformatics	01/2014 – 08/2018
Swiss Institute of Bioinformatics (SIB), University of Basel, Basel, Switzerland	Visiting Research Scientists	Computational Biology	06/2009 – 08/2009
Swiss Federal Institute of Technology Zurich (ETH Zurich), Basel, Switzerland	Visiting Research Scientists and ThinkSwiss Fellow	Computational Genomics	06/2008 – 08/2008

3.3. Entrepreneurial Experience

Place of Employment	Title	Field	Dates (Month &Year)
AI-NeoTech LLC	Founder	Neuroscience Tech	08/2021 to date

3.4. Teaching Experience

3.4.1. Courses Developed and/or significantly revised

Florida International University (FIU)
Knight Foundation School of Computing and Information Sciences, Miami, FL, USA

1. CDA 3102, Computer Architecture (undergraduate course)
2. CAP 2752 Fundamentals of Data Science (undergraduate course)
3. CIS 6930 HPC Algorithms for Scientific Applications (graduate course)
4. COT 4431/COT 5432 Applied Parallel Computing (cross listed undergrad/grade course)
5. CIS 4930/5931, Special Topics course, Computational Methods for Biomedical Data (cross listed undergrad/grade course)
6. CAI 4775, Advanced Computational Methods in Health & Biomedical Data
7. CAI 5777, Advanced Algorithmic and Machine-Learning for Health & Biomedical Data

Western Michigan University (WMU)
Department of Computer Science, Kalamazoo, MI, USA
CS6030/ECE6950, High Performance Architectures & Algorithms for Big Data

3.4.2. Courses Taught

Florida International University (FIU)
Knight Foundation School of Computing and Information Sciences, Miami, FL, USA

Course	Semester
CIS 4930/5931, Special Topics course, Computational Methods for Biomedical Data	Fall 2025
COT 4431/COT 5432, Applied Parallel Computing	Fall 2025
CDA 3102, Computer Architecture	Fall 2024
COT 4431/COT 5432, Applied Parallel Computing	Fall 2024
CDA 3102, Computer Architecture	Fall 2023
COT 4431/COT 5432, Applied Parallel Computing	Fall 2023
COP 3337, Programming II	Spring 2023
CDA 3102, Computer Architecture	Fall 2022
COT 4431/COT 5432, Applied Parallel Computing	Fall 2022

COP 3337, Programming II	Spring 2022
CDA 3102, Computer Architecture	Fall 2021
COT 4431/COT 5432, Applied Parallel Computing	Fall 2021
COP 3337, Programming II	Spring 2021
COP 3337, Programming II	Fall 2020
COT 4431/COT 5432, Applied Parallel Computing	Fall 2020
COP 3337, Programming II	Spring 2020
COP 3337, Programming II	Fall 2019
CDA 3103, Fundamentals of Computer Systems	Spring 2019
CDA 3103, Fundamentals of Computer Systems	Fall 2018

Western Michigan University
Department of Computer Science, and Department of Electrical & Computer Engineering, Kalamazoo, MI, USA

Course	**Semester**
CS 4310, Design & Analysis of Algorithms	Spring 2018
ECE 2510, Introduction to Microprocessors	Fall 2017
ECE 2510, Introduction to Microprocessors	Spring 2017
CS6030/ECE6950, High Performance Architectures & Algorithms for Big Data	Spring 2017
ECE 2510, Introduction to Microprocessors	Fall 2016
CS 4310, Design & Analysis of Algorithms	Spring 2016
ECE 2510, Introduction to Microprocessors	Fall 2015
CS6030/ECE6950, High Performance Architectures & Algorithms for Big Data	Spring 2015
ECE 2510, Introduction to Microprocessors	Fall 2014
ECE/CS 7250, Doctoral Research Seminar	Spring 2014

National Institutes of Health (NIH)
Systems Biology Center, Bethesda, MD, USA

Course	**Semester**
Instructor for Community College Summer Enrichment Program (CCSEP)	Summer 2012

University of Illinois at Chicago (UIC)
Electrical & Computer Engineering, Chicago, IL, USA

Course	**Semester**
Digital Communications Course (ECE 432)	Fall 2007
Teaching Assistant for Electric Circuit Analysis Lab (ECE 225)	Spring 2007

3.4.3. Course Certifications

Quality Matters (QM) Certification for COP 3337, Programming II — Fall 2020

4. Honors and Awards

1. **Associate Editor***, IEEE Transactions on Computational Biology and Bioinformatics (TCBB),* 2025
2. *Excellence in Research and Creative Activities Award*, Knight Foundation School of Computing and Information Science (KFSCIS), Florida International University (FIU), Dec 2024 (1 faculty each year is selected)
3. **Editor**, *PLOS Mental Health*, Oct 2023
4. ***FIU Top Scholar*** Award (Research and Creative Activities), 2022
5. **Keynote Speaker** at the 14th International Conference on Bioinformatics and Computational Biology (BICOB), Feb 2022
6. **Distinguished Speaker** at the 15th IEEE International Conference on Open-Source Systems & Technologies (ICOSST), 2021
7. *Excellence in Applied Research Award*, Knight Foundation School of Computing and Information

Science (SCIS), Florida International University (FIU), Dec 2020 (1 faculty each year is selected)
8. **NSF CAREER Award**, 2017-2022
9. **Board member** for Statistics and Bioinformatics section, Journal of the American Society of Nephrology (JASN), Feb 2019 to date
10. *Distinguished Research and Creative Scholarship Award*, Office of Vice President of Research WMU, Feb 2018
11. *Highest Funding Award* in a year for computer scientist, WMU 2017
12. **ACM Senior Member**, May 2017
13. *Outstanding New Researcher Award*, College of Engineering and Applied Science (CEAS), Western Michigan University, Jan 2016 (1 faculty member gets the award in a single year for the entire college consisting of 7 academic departments)
14. **IEEE Senior Member**, March 2015
15. *NSF CISE Research Initiation Initiative (CRII) Award*, Feb 2015 - Feb 2018
16. *Fellows Award for Research Excellence (FARE)*, National Institutes of Health (NIH), June 2012 (Official award ceremony and US$1000 travel grant)
17. *Postdoctoral Fellowship Award*, National Institutes of Health (NIH), 2010-2011.
18. *Travel Award* from Swiss Institute of Bioinformatics (SIB), Summers 2009.
19. *UIC Graduate Student Council (GSC) travel award* for the BiCoB April 2009.
20. *Think Swiss Scholarship*, by the Government of Switzerland for two years (2007 and 2008).
21. *Travel award* from D-BSSE ETH Zurich, Summers 2008.
22. Higher Education Commission Scholarship (≈ US$ 100,000) by Government of Pakistan, 2007 (13/800 =1.63% acceptance rate in engineering discipline).

5. Research Funding

Since 2015 I have been awarded over **US$ 6.85 million** in external research funds - with more than **US$ 5.45 million** as a PI. External fund raising (**US$ 10 Million** + **US$ 6 Million**), Intramural funds (**US$ 895,107**) and computing allocations are not included in this amount.

5.1. Funded External Grants

1. **National Science Foundation (NSF) IIS-2530255 [US$ 600,000]**, "Collaborative Research: CISE-ANR:III: Small: Leveraging External Data for Enhanced Understanding and Causal Attribution of Anomalies in Wastewater Networks", **Fahad Saeed (PI)**, with Kamal Premaratne (univ. Of Miami) and Helena Solo-Gabriele (Univ. of Miami), Benferhat Salem (Artois University, France), DELENNE Carole (Université de Montpellier France), and Nanee Chahinian (Aix Marseille University (Polytech Marseille/IUSTI)), Feb 2026 - Jan 2029 (combined funding of **US$1.2 Million with 600k from French ANR, and 600k from NSF**)
2. **National Institutes of Health (NIH) R35GM153434 [US$ 1.75 million]**, "Machine-Learning Models for big data omics", **Fahad Saeed (PI)**, June 2024 - June 2029 (single PI grant: MIRA R35 Outstanding Investigator mechanism)
3. **National Science Foundation (NSF) TIP-2322346 [US$ 275,000]**, "SBIR Phase I: STTR Phase I: Patient-Specific System for Early Detection and Identification of Epileptic Seizures", Saba Mehmood (PI), **Fahad Saeed (Co-PI)**[3], Oct 2023 to Oct 2024
4. **National Science Foundation (NSF) OAC-2312599 [US$ 600,000]**, "OAC Core: High Performance Computing Algorithms and Software for large-scale Mass Spectrometry based Omics", **Fahad Saeed (PI)**, Sept 2023 - August 2026
5. **National Science Foundation (NSF) CHE-2304837 [US$ 500,000]**, "Development of Multi-dimensional Ion Mobility-Tandem Mass Spectrometry (IMSn-FT-ICR MSn) Tools for the Characterization of Complex Mixtures", Francisco Fernandez-Lima(PI), **Fahad Saeed (Co-PI)**, Sept 2023 - August 2026
6. **National Science Foundation (NSF) TI-2213951 [US$ 250,000]**, "PFI-TT: Artificial Intelligence-enabled Real-time System for Early Epileptic Seizure Detection and Prediction", **Fahad Saeed (PI)**, August 1, 2022 - July 31, 2025
7. **National Science Foundation (NSF) IIP-2143515 [US$ 50,000]**, "I-Corps: Utilizing Machine

[3] I was PI on this grant but NSF rules does not allow me to be PI while at FIU for a company that I founded - so my mentee Ms. Mehmood was substituted as PI

learning and Artificial Intelligence (AI) for Early Detection and Identification of Mental Disorders", **Fahad Saeed (PI)**, Sept 2021 - Sept 2023

8. **National Science Foundation (NSF) OAC-1925960 [US$ 415,590]**, "CAREER: Towards Fast and Scalable Algorithms for Big Proteogenomics Data Analytics", **Fahad Saeed (PI)**, Sept 2018 - August 2023
9. **National Science Foundation (NSF) OAC-2126253 [US$ 400,000]**, "CC* Compute: RAP- TOR - Reconfigurable Advanced Platform for Transdisciplinary Open Research", Jason Liu (PI), Jayantha Obeysekera (Co-PI), Keqi Zhang (Co-PI), Cassian D'Cunha (Co-PI), Mike Kir- gan (SI), Yuepeng Li (SI), Vasilka Chergarova (SI), Yagya Joshi (SI), Jonathan Casco (SI), and **Fahad Saeed (Co-PI)**, Sept 2021 - Sept 2023
10. **National Institutes of Health (NIH) Supplemental- 3R01GM134384-02S1 [US$ 100,000]**, "Compute-Cluster for Deep-Learning Models for Mass Spectrometry based Proteomics data", **Fahad Saeed (PI)**, August 2021 - May 2024
11. **National Institutes of Health (NIH) Supplemental- 3R01GM134384-01A1S1 [US$ 205,291]**, "Multimodal Machine-Learning Algorithms for Early Detection, and Classification for Alzheimer Disorder and Related Dementia's", **Fahad Saeed (PI)**, May 2021 - May 2024
12. **National Institutes of Health (NIH) R01GM134384 [US$ 965,874]**, "Multimodal Machine-Learning and High-Performance Computing Strategies for Big MS Proteomics Data", **Fahad Saeed (PI)** with Shu-Ching Chen (Co-Investigator), Jason Liu (Co-Investigator), Francisco Alberto Fernandez-Lima (Co-Investigator), and Sitharama Iyengar (Senior Personal), June 2020 - June 2024
13. **National Science Foundation (NSF) CCF-1855441 [US$ 7,708]**, "CRII: SHF: HPC Solutions to Big NGS Data Compression", **Fahad Saeed (PI)**, Sept 2018 - Jan 31, 2020
14. **National Science Foundation (NSF) ACI-1651724 [US$ 499,999]**, "CAREER: Towards Fast and Scalable Algorithms for Big Proteogenomics Data Analytics", **Fahad Saeed (PI)**, April 2017-April 2022
15. **National Institutes of Health (NIH) R15GM120820 [US$ 418,533]**, "Parallel Algorithms for Big Data from Mass Spectrometry based Proteomics", **Fahad Saeed (PI)**, April 2017 - April 2020
16. **National Science Foundation (NSF) REU Supplement [US$ 16,000]**, "CRII: SHF: HPC Solutions to Big NGS Data Compression", **Fahad Saeed (PI)**, Feb 2016 - Feb 2018
17. **National Science Foundation (NSF) CCF-1464268 [US$ 171,341]**, "CRII: SHF: HPC Solutions to Big NGS Data Compression", **Fahad Saeed (PI)**, (Feb 2015 - Feb 2018)
18. **National Science Foundation (NSF) CNS-1250264 [US$ 200,000]**, "EAGER: High Performance Algorithms and Implementations for Biological Sequence Analysis and Genome Alignment", Ashfaq Khokhar, **Fahad Saeed (Co-PI)** (Sept 2012 - Aug 2015)

5.2. Funded Equipment/Computing Allocations

1. **NSF ACCESS Super Computing Infrastructure [ACCESS Credits: 2,000,000]**, "Continued Development of scalable Machine-learning models for Scientific data", **Fahad Saeed (PI)**, (July 2023 - June 2024)
2. **OpenBCI [US$ 2,858]**, "Real-time edge-based seizure prediction using EEG data", Saba Mehmood (PI), Fahad Saeed (Senior Personal), Umair Muhammad (Senior Personal), Equipment grant: Cyton + Daisy Biosensing Boards (16-Channels), Header Pin to Touch Proof Electrode Adapter, Gelfree BCI Electrode Cap Kit), August 2023
3. **Xilinx [US$ 13,195]**, "Design and development of FPGA based MS omics pipeline", **Fahad Saeed(PI)** (Equipment Grant, Versal AI Core EK-VCK190-G FPGA), March 2022
4. **NSF XSEDE Extended Collaborative Support Service (ECSS) [US$ 50,000]**, "DeepSNAP: Scalable Machine Learning for Mass Spectrometry based Proteomics", **Fahad Saeed (PI)**, (Jan 2021 - Dec 2021)
5. **National Science Foundation XSEDE ASC200004 [125,000.0 Service Units (SU)/10,000.0 GB SDSC Medium-term disk storage (Data Oasis)/10,000.0 GPU Hours/75,000 Core hours on Clusters: US$ 43,161.42]**, "DeepSNAP: Scalable Machine Learning for Mass Spectrometry based Proteomics", **Fahad Saeed (PI)**, (Jan 2021 - Dec 2021)
6. **National Science Foundation XSEDE ASC200004 [100,000.0 Service Units (SU)/10,000.0 GB SDSC Medium-term disk storage (Data Oasis)/5,000.0 GPU Hours: US$ 6,190]**, "DeepSNAP: Scalable Machine Learning for Mass Spectrometry based Proteomics", **Fahad Saeed (PI)**, (March

2020 - Jan 2021)

7. **Intel Altera [US$ 7,900]**, “MS proteomics analysis using reconfigurable hardware”, **Fahad Saeed (PI)** (Equipment Grant, DE10-PRO-SX FPGA), Nov 2019
8. **National Science Foundation XSEDE supplemental grant TG-CCR150017 [30,000 Ser- vice Units (SU)/6TB SDSC Disk Storage/2500 GPU Hours: US$ 450]**, “Smart Index and Search for De Novo Proteogenomics”, **Fahad Saeed (PI)**, (March 2019 - June 2020)
9. **National Science Foundation XSEDE renewal grant TG-CCR150017 [30,000 Service Units (SU)/6TB SDSC Disk Storage/2500 GPU Hours: US$ 3,159]**, “Smart Index and Search for De Novo Proteogenomics”, **Fahad Saeed (PI)**, (March 2019 - March 2020)
10. **NVIDIA [US$ 1149]**, “High Performance Algorithms for Big Data Proteomics”, **Fahad Saeed (PI)** (Equipment Grant for NVIDIA TITAN Xp GPU), August 2018
11. **National Science Foundation XSEDE renewal grant TG-CCR150017 [30,000 Service Units (SU)/6TB SDSC Disk Storage: US$ 6564]**, “A Distributed-Shared Memory Strategy to Speedup the Compression of Big Next-Generation Sequencing Datasets”, **Fahad Saeed (PI)**, (June 2016 - June 2018)
12. **National Science Foundation XSEDE startup grant TG-CCR150017 [30,000 Service Units (SU)]**, “Scalability study of compression algorithms for peta scale NGS data”, **Fahad Saeed (PI)**, (June 2015 - June 2016)
13. **Intel Altera [US$ 16,000]**, “Short Reads mapping to the genome using reconfigurable hard- ware”, **Fahad Saeed (PI)** (Equipment Grant, 2 DE5-NET-450 FPGA’s), April 2014
14. **NVIDIA [US$ 5499]**, “High Performance Algorithms for Genome Alignments”, **Fahad Saeed (PI)** (Equipment Grant for Tesla K40 GPU), Feb 2014

5.3. Funded Intramural Grants

1. **FIU Population Health Initiative (PHI) Research Catalyst Pilot Program [US$ 50,000]**, “An Integrated Multi-Modal Framework to Quantify Life-Course Dementia Risk and Preventable Burden in U.S. Populations”, Natalia Trujillo Orrego (PI), **Fahad Saeed (Co-PI),** Feb 2026 - Feb 2027
2. **University of Florida ORED/BOG [US$ 50,000]**, “Scalable Algorithms for Big MS data using CPU-GPU architectures”, **Fahad Saeed (PI)**, October 18, 2021 - April 18th, 2022,
3. **Office of Vice President of Research, Florida International University (FIU) [US$ 435,281]**, “Scalable Algorithms for Big Omics Data”, **Fahad Saeed (PI)**, Sept 2018 - April 2022
4. **Office of Vice President of Research, Western Michigan University (WMU) [US$ 129,570]**, “Scalable Algorithms for Big Proteogenomics Data Analytics”, **Fahad Saeed (PI)**, April 2017 - April 2020
5. **College and Engineering and Applied Science (CEAS), Western Michigan University (WMU) [US$ 41,650]**, “Developing HPC solutions to big fMRI data”, **Fahad Saeed (PI)**, April 2017 - June 2018
6. **Department of Computer Science, Western Michigan University (WMU) [US$ 238,606]**, “High performance computing for Computational Biology”, **Fahad Saeed (PI)**, Jan 2014 - June 2018

5.4. External Fund Raising

1. Nicolai Kosche Professor of Computer Science at Western Michigan University - Spear- headed the proposal writing and execution of Bioinformatics, and Machine-learning models for Personalized Medicine Center which resulted in chaired faculty positions in department of biology, medicine, and computer science - Total **US$ 6 Million** - 2020 (bequest)
2. FIU Knight Foundation Grant- Collaborative proposal writing with SS Iyengar, Robert Lipman, Mark Finlayson, Joselyn Naranjo, Scott Graham, Jason Liu and John Volakis - Resulted in renaming of school to **Knight School of Computing and Information Sciences** - Total **US$ 10 Million** to the School of Computing FIU - 2021

5.5. Pending

1. **National Institutes of Health (NIH) [US$ 2,025,213]**, “Self-Supervised Multimodal Models for the Early Prediction of Alzheimer's Disease”, **Fahad Saeed (PI),** Aviva Abosch, Emma Lovejoy Ducca, Trujillo, Natalia (Co-I), June 2026
2. **National Institutes of Health (NIH) [US$ 709,691]**, “HPC & GPU Resources for ML Biomedical

Research at FIU", **Fahad Saeed (PI)**, Prem Chapagain (Co-I), Joshua Hutcheson (Co-I), Janna Fierst (Co-I), Boubakari Ibrahimou (Co-I), Julio Ibarra (Advisory), Jorge Diaz (Advisory), Jason Liu (Advisory), Mariana Sanchez (Co-I), Ananda Mondal (Co-I), Xuyu Wang (Co-I), Kaoutar Ben Ahmed (Co-I), Mohammad Rahman (Co-I), Anthony McGoron (Co-I), June 2026

3. **Environmental Protection Agency (EPA) [US$ 174,822],** "Fuzzy Cognitive Mapping of Watershed-to-Reef Connectivity in the Biscayne Coastal Ecosystem", **Fahad Saeed (Co-PI)**, Oliver Adigun (PI), March 2026
4. **National Institutes of Health (NIH) [US$ 286,009]**, "Self-Learning models for diagnosis and early prediction of Alzheimer's disease ", **Fahad Saeed (PI)**, Trujillo, Natalia (Co-I), March 2026
5. **National Science Foundation (NSF) [US$ 390,000]**, "CICI:TCR:Safeguarding Scientific AI Workflows Within Shared Computational Infrastructures", **Fahad Saeed (Co-PI)**, Amin Kharraz (PI), Yanzhao Wu (Co-PI), Jan 2026
6. **FIU PHI Research Catalyst Pilot Fund [US$ 50,000]**, "An Integrated Multi-Modal Framework to Quantify Life-Course Dementia Risk and Preventable Burden in U.S. Populations", **Fahad Saeed (PI)**, Trujillo, Natalia (PI), Byomkesh Talukder (Co-PI), Weize Wang (Co-PI), Jan 2026
7. **Florida Department of Health (FDOH) [US$ 335,109]**, "Multimodal Machine Learning Guided Dosimetry to Predict and Improve Response to I-131 Therapy in Papillary Thyroid Carcinoma", **Fahad Saeed (Co-PI)**, Anthony McGoron (PI), Jan 2026
8. **Florida Department of Health (FDOH) [US$ 100,000]**, "Alzheimer screening Using Ocular Data and Machine-Learning Models", **Fahad Saeed (PI)**, Shuliang Jiao (Co-I) and Trujillo, Natalia (Co-I), Nov 2025
9. **Chan Zuckerberg Foundation [US$ 500,000]**, "Cell-specific extracellular vesicle communication in thyroid carcinoma", Anthony McGoron (PI), **Fahad Saeed (Co-PI),** Nov 2025
10. **National Institutes of Health (NIH) [US$ 286,009]**, "Predicting Alzheimer's Using Ocular Data and Machine-Learning Models ", **Fahad Saeed (PI)**, Shuliang Jiao (Co-I) and Trujillo, Natalia (Co-I), Oct 2025
11. **National Science Foundation (NSF) [US$ 1,250,000]**, "STTR Phase II: Patient-Specific System for Early Detection and Identification of Epileptic Seizures", **Fahad Saeed (Co-PI)**, Saba Mehmood (PI), Sept 2025
12. **National Science Foundation (NSF) [US$ 1,200,000]**, "NSF TTP-T: Scalable Cross-Platform Software Prototype integrating HPC, and ML models for Mass Spectrometry based omics data", **Fahad Saeed (PI)**, Francisco Lima (Co-PI), Sept 2025
13. **National Institutes of Health (NIH) [US$ 285,876]**, "Generalizable machine learning models for early prediction of Alzheimer's disease in diverse demographic populations", **Fahad Saeed (PI)**, June 2025
14. **National Institutes of Health (NIH) [US$ 2,148,445]**, "Multimodal Machine Learning models for seizure prediction", **Fahad Saeed (PI)** with Christian Poellabauer, Oleksii Shandra, and George A. Buzzell June 2025
15. **National Science Foundation (NSF) [US$600,000]**, "Collaborative Research: CISE-ANR:III: Small: Leveraging External Data for Enhanced Understanding and Causal Attribution of Anomalies in Wastewater Networks", **Fahad Saeed (PI for US Side)**, Kamal Premaratne (Univ. of Miami), Helena Solo-Gabriele (University of Miami), DELENNE Carole (AMU France), Nanee Chahinian (HydroSciences Montpellier, Univ. Montpellier (France)), Salem Benferhat (CRIL, University of Artois (France)), March 2025
16. **National Institutes of Health (NIH) [US$ 3,450,460]**, "Generalizable machine learning models for early prediction of Alzheimer's disease in diverse demographic populations", **Fahad Saeed (PI)** with Serdar Bozdag (MPI, UNT), Robert C. Barber (UNTHC), Melissa Petersen (UNTHC) and Angela Laird (FIU), Feb 2025
17. **National Institutes of Health (NIH) [US$ 240,000]**, "GPU machines to train and validate Large Language Models for Mass Spectrometry based omics data", **Fahad Saeed (PI)**, Feb 2025
18. **National Science Foundation (NSF) [US$974,690]**, "PDaSP: Track1: Practically Deployable Multi-Party Computation for Privacy-preserving Machine Learning in Health Applications", Kemal Akkaya (PI), **Fahad Saeed (Co-PI)**, Farhad Shirani Chaharsooghi (Co-PI), April 2024

7. Publications and Creative Scholarship

† Indicates Corresponding and/or Senior Author.
Impact factor, and conference acceptance rate is listed, if available.

7.1. Books

1. Muhammad Haseeb, and **Fahad Saeed**†, "High Performance Computing for Big Mass Spectrometry Data based omics", ISBN: 978-3-031-01959-3, Hardcover Edition, *Springer*, Sept 2022 (140 pages)

7.2. Book Chapters

1. Umair Mohammad and **Fahad Saeed**†, "Heterogeneity Aware Distributed Machine Learning at the Wireless Edge for Health IoT Applications: An EEG Data Case Study", accepted in the *Springer Distributed Optimization and ML*, August 2023.
2. Usman Tariq, Samuel Ebert, and **Fahad Saeed**†, "Making MS omics data ML ready: SpeCol- late Protocols", Accepted in *Nature Springer Methods in Molecular Biology*, August 2023
3. Taban Eslami, Joe Raiker, and **Fahad Saeed**†, "Explainable and Scalable Machine-Learning Algorithms For Detection of Autism Spectrum Disorder using fMRI Data", *Elsevier Neural Engineering Techniques for Autism Spectrum Disorder Volume 1: Imaging and Signal Analysis*, Pages 39-54 (Chapter 4), 2021

7.3. Edited Conference Proceedings

1. Proceedings of 7th International Conference on Bioinformatics and Computational Biology (BICoB), with Hisham Al-Mubaid and Nurit Haspel, (ISBN: 9781510800137) March 2015
2. Proceedings of 6th International Conference on Bioinformatics and Computational Biology (BICoB), with Bhaskar Dasgupta, Hisham Al-Mubaid and Nurit Haspel (ISBN: 9781632665140) March 2014
3. Proceedings of 5th International Conference on Bioinformatics and Computational Biology (BICoB), with Bhaskar Dasgupta, and Hisham Al-Mubaid (ISBN: 978-1-880843-89-5), March 2013
4. Proceedings of 4th International Conference on Bioinformatics and Computational Biology (BICoB), with Hisham Al-Mubaid and Ashfaq Khokhar (ISBN: 978-1-880843-85-7), March 2012

7.4. Edited Special Issue of Journals

1. Special issue on selected papers from the 4th IEEE HPC-BOD workshop, with Serdar Bozdag in the *Frontiers of Bioinformatics*, 2024
2. Special issue on selected papers from the 3th IEEE HPC-BOD workshop, with Serdar Bozdag in the *PLos One*, 2022 (due to small number of papers - the selected manuscripts were published as regular papers).
3. Special issue on selected papers from the 7th international conference on bioinformatics and computational biology (BICoB 2015) with Nurit Haspel and Hisham Al-Mubaid in the *Journal of Bioinformatics and Computational Biology (JBCB)* Vol. 14, No. 3. March 2016
4. Special issue on selected papers from the 6th international conference on bioinformatics and computational biology (BICoB 2014) with Bhaskar Dasgupta, Nurit Haspel and Hisham Al- Mubaid in the *Journal of Bioinformatics and Computational Biology (JBCB)* Volume 12, Issue 05, October 2014
5. Special issue on selected papers from the 5th international conference on bioinformatics and computational biology (BICoB 2013) with Bhaskar Dasgupta and Hisham Al-Mubaid, *Journal of Bioinformatics and Computational Biology (JBCB)* Volume 11, Issue 05, October 2013

7.5. Peer-Reviewed Journal Publications

1. Hampson CL, Peraza JA, Guerrero LM, Bottenhorn KL, Riedel MC, Almuquin F, Smith DD, Schmarder KM, Crooks KE, Sangoi JA, Keller KR, Pintos Lobo R, Sutherland MT, Musser ED, Dai Y, Agarwal R, **Fahad Saeed**†, Angie Laird. "Habenula alterations in resting state functional connectivity among autistic individuals". *Elsevier Biol Psychiatry: Cogn Neurosci Neuroimag*, In Press. 2026 (Impact factor: 4.7)
2. Yashu Vashishath, Bizhan Alipour Pijani, Neha Goud Baddam, **Fahad Saeed**†, Serdar Bozdag, Alzheimer's Disease Neuroimaging Initiative, Predicting progression of Alzheimer's disease using blood-based multi-omics data, article vbag085, *Oxford Bioinformatics Advances*, March 2026
3. Usman Tariq, Bilal Shabbir, and **Fahad Saeed**†, "End-to-end deep attention-based multitask pipeline for predicting uncertainty-quantified peptide properties from mass spectrometry data",

Nature Scientific Reports, March 2026 (Impact Factor: 3.9)

4. Umair Mohammad, and **Fahad Saeed**†, "MLSPred-Bench: Transforming Electroencephalography (EEG) Datasets into Machine Learning-Ready Seizure Prediction Benchmarks, Vol. 15, pp. 103574, *Elsevier MethodsX*, 2025 (Impact Factor: 1.9)
5. Fahad Almuqhim, and **Fahad Saeed**†, "Overcoming Site Variability in Multisite fMRI Studies: An Autoencoder Framework for Enhanced Generalizability of Machine Learning Models", Vol. 23, No. 3, pp. 1-19, *Springer Nature Neuroinformatics*, 2025 (Impact Factor: 3.1)
6. Maryam Akhavan Aghdam, Serdar Bozdag, **Fahad Saeed**†, "Machine-learning Models for Alzheimer's Disease Diagnosis using Neuro Imaging data: Survey, Reproducibility, and Generalizability Evaluation", *Springer Nature Brain Informatics*, Vol 12, article 8, Issue 1, Pages 1-27, March 2025 (Impact Factor: 4.5)
7. Muhammad Haseeb, and **Fahad Saeed**†, "GPU-Acceleration of the Distributed-Memory Database Peptide Search of Mass Spectrometry Data", *Nature Scientific Reports*, Vol. 13, Article 18713, 31st Oct 2023 **(Impact Factor: 4.9)**
8. Mohammad Al Olaimat, Jared Martinez, **Fahad Saeed**, Serdar Bozdag, "PPAD: A deep learning architecture to predict progression of Alzheimer's disease", *Oxford Bioinformatics*, Volume 39, Issue Supplement 1, Pages i149–i157, June 2023 **(Impact Factor: 5.8)**
9. Oswaldo Artiles, Zeina Al Masry, and **Fahad Saeed**†, "Confounding effects on the performance of machine learning analysis of static functional connectivity computed from rs-fMRI multi-site data", *Springer Neuroinformatics*, Vol 21, pp. 651-668, August 2023 **(Impact Factor: 4.0)**
10. Leyva, Dennys, Usman Tariq, Muhammad, Jaffe, Rudolf, **Fahad Saeed**, Francisco Fernández-Lima, "Description of Dissolved Organic Matter transformational networks at the molecular level", *ACS Journal of Environmental Science & Technology*, Vol 57, No. 6, pages 2672- 2681, Jan 2023 **(Impact Factor: 11.4)**
11. Dennys Leyva, Rudolf Jaffé, Jessica Courson, John S. Kominoski, Muhammad Usman Tariq, **Fahad Saeed**, and Francisco Fernandez, "DOM along a freshwater-to-estuarine coastal gradient in the Florida Everglades", *Springer Aquatic Sciences*, volume 84, Article number: 63, Sept 2022 **(Impact Factor: 2.8)**
12. Mohammed Aledhari, Rehma Razzak, Basheer Qolomany, Ala Al-Fuqaha, and **Fahad Saeed**†, "Biomedical IoT: Enabling Technologies, Architectural Elements, Challenges, and Future Directions", vol. 10, pp. 31306-31339, *IEEE Access*, March 2022 **(Impact Factor: 3.4)**
13. Dennys Leyva, Muhammad Usman Tariq, Rudolf Jaffe, **Fahad Saeed**, and Francisco Fernández-Lima, "Unsupervised Structural Classification of Dissolved Organic Matter based on frag- mentation pathways", *ACS Journal of Environmental Science & Technology*, Vol 56 (2), pp 1458-1468, Jan 2022 **(Impact Factor: 9.1)**
14. **Fahad Saeed**†, Muhammad Haseeb, and SS Iyengar, "Communication lower-bounds for distributed-memory computations for mass spectrometry-based omics data", *Journal of Parallel and Distributed Computing (JPDC)*, Volume 161, Pages 37-47, March 2022 **(Impact Factor: 3.7)**
15. Usman Tariq, and **Fahad Saeed**†, "SpeCollate: Deep cross-modal similarity network for mass spectrometry data-based peptide deductions", *PLoS ONE*, Vol. 16, Issue 10, Oct 2021 **(Im- pact Factor: 3.24)**
16. Muhammad Haseeb, and **Fahad Saeed**†, "High performance computing framework for tera- scale database search of mass spectrometry data", ***Nature Computational Science***, Vol. 1, Issue 8, p. 550–561, August 2021 (**Impact Factor: 18.3**)
17. Fahad Almuqhim, and **Fahad Saeed**†, "ASD-SAENet: Sparse Autoencoder for detecting autism spectrum disorder (ASD) using fMRI data", *Frontiers in Computational Neuroscience*, Vol. 15, p. 27, March 2021 **(Impact Factor: 3.4)**
18. Muaaz Awan, and **Fahad Saeed**†, "Benchmarking Mass Spectrometry based Proteomics Algorithms using a Simulated Database", *Springer Network Modeling Analysis in Health Informatics and Bioinformatics*, Vol. 10, Article 23, March 2021 **(Impact Factor: 2.3)**
19. Taban Eslami, Fahad Almuqhim, Joseph S Raiker, **Fahad Saeed**†, "Machine Learning methods for diagnosing Autism Spectrum Disorder, and Attention-deficit/Hyperactivity Disorder using functional and structural MRI: A Survey", *Frontiers in Neuroinformatics*, Vol. 14, pp. 62, Jan 2021 **(Impact Factor: 2.7)**
20. Usman Tariq, Muhammad Haseeb, Mohammed Aledhari, Rehma Razzak, Reza Parizi, and **Fahad Saeed**†, "Methods for Proteogenomics Data Analysis, Challenges, and Scalability Bottlenecks:

A Survey", Vol. 9, pages: 5497-5516, *IEEE Access*[4], Jan 2021 **(Impact Factor: 4.1)**

21. Bronte Wen, Hyun Jun Jung, Lihe Chen, **Fahad Saeed**, and Mark Knepper, "NGS-Integrator: An efficient tool for combining multiple NGS data tracks using minimum Bayes' factors", *BMC Genomics*, vol. 21, No. 1, pp-1-7, Nov 2020 **(Impact Factor: 4.09)**

22. Mohammed Aledhari, Rehma Razzak, Reza Parizi, and **Fahad Saeed**†, "Federated Learning: A Survey on Enabling Technologies, Protocols, and Applications", *IEEE Access*, Vol. 8, pp. 140699–140725, July 2020 **(Impact Factor: 4.1)**

23. Taban Eslami, Vahid Mirjalili, Alvis Fong, Angela Laird, and **Fahad Saeed**†, "ASD-DiagNet: A hybrid learning approach for detection of Autism Spectrum Disorder using fMRI data", *Fron- tiers In Neuroinformatics*, Vol 13, Pages 70, Nov 2019 **(Impact Factor: 3.8)** **(Best Paper Editor's Pick 2021 - one of 23 papers selected from over 700 papers)**[5]

24. Muaaz Awan, and **Fahad Saeed**†, "MaSS-Simulator: A highly configurable simulator for generating MS/MS datasets for benchmarking of proteomics algorithms", *Wiley PROTEOMICS*[6], Volume 18, Issue 20, October 2018 **(Impact Factor: 4.1)**

25. Muaaz Awan, Taban Eslami, and **Fahad Saeed**†, "GPU-DAEMON: GPU Algorithm Design, Data Management & Optimization template for array based big omics data", *Elsevier Computers in Biology and Medicine*, Volume 101, pp. 163-173, October 2018 **(Impact Factor: 2.2)**

26. **Fahad Saeed**†, "Towards quantifying psychiatric diagnosis using machine learning algorithms and big fMRI data", *BMC Big Data Analytics*, Vol. 3, No. 1, pp. 1-7, May 2018

27. Taban Eslami, and **Fahad Saeed**†, "Fast-GPU-PCC: A GPU-Based Technique to Compute Pairwise Pearson's Correlation Coefficients for Time Series Data - An fMRI Study", *MDPI High-Throughput*, Vol. 7, No. 2, article 11, April 2018

28. Mohammed Aledhari, Marianne E Di Pierro, Mohamed Hefeida, and **Fahad Saeed**†, "A Deep Learning-Based Data Minimization Algorithm for Fast and Secure Transfer of Big Genomic Datasets", *IEEE Transactions on Big Data*, Feb 2018 **(Impact Factor: 7.2)**

29. Sandino N. Vargas-Pérez and **Fahad Saeed**†, "A Hybrid MPI-OpenMP Strategy to Speedup the Compression of Big Next-Generation Sequencing Datasets", *IEEE Transactions on Parallel and Distributed Systems*[7], vol. 28, no. 10, pp. 2760-2769, Oct. 1 2017 **(Impact Factor: 4.2)**

30. Pablo C. Sandoval, J'Neka S. Claxton, Jae Wook Lee, **Fahad Saeed**, Jason D. Hoffert and Mark A. Knepper, "Systems-level analysis reveals selective regulation of Aqp2 gene expression by vasopressin", *Nature Scientific Reports*, Vol. 6, article number 34863, October 2016 **(Impact Factor: 5.6)**

31. Muaaz Gul Awan and **Fahad Saeed**†, "MS-REDUCE: An ultrafast technique for reduction of Big Mass Spectrometry Data for high-throughput processing", *Oxford Bioinformatics*[8], Vol. 32, No. 10, pages 1518-1526, Jan 2016 **(Impact Factor: 7.3)**

32. Sookkasem Khositseth, Panapat Uawithya, Poorichaya Somparn, Komgrid Charngkaew, Nattakan Thippamom, Jason D. Hoffert, **Fahad Saeed**, D. Michael Payne, Shu Hui Chen, Robert A. Fenton and Trairak Pisitkun, "Autophagic degradation of aquaporin-2 is an early event in hypokalemia-induced nephrogenic diabetes insipidus", *Nature Scientific Reports*, Vol. 5, article number 18311 Dec 2015 **(Impact Factor: 5.6)**

33. Akshay Sanghi, Matthew Zaringhalam, Callan Corcoran, **Fahad Saeed**, Jason Hoffert, Pablo Sandoval, Trairak Pisitkun, and Mark Knepper, "A Knowledge Base of Vasopressin Actions in Kidney", *American Journal of Physiology (AJP)*, Vol. 307, No. 6, pages F747–F755, July 2014 **(Impact Factor: 7.6)**

34. **Fahad Saeed**†, Jason Hoffert, Trairak Pisitkun and Mark Knepper, "Exploiting Thread-Level and

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[4] IEEE Access is ranked as one of the top 3 peer-reviewed journals in its respective sub-category https://scholar.google.com/citations?view_op=top_venues&hl=en&vq=eng_enggeneral

[5] Editor citation reads: *This collection showcases well-received spontaneous articles from the past 2 years. All re- search presented demonstrates strong advances in theory, experiment and methodology with applications to compelling problems. This collection aims to recognize highly deserving authors*.

[6] Wiley Proteomics is one of the top 5 peer-reviewed journal in its respective sub-category https://scholar.google.com/citations?view_op=top_venues&hl=en&vq=bio_proteomicspeptides

[7] IEEE Transactions on Parallel and Distributed Systems is considered *the* top (highest ranked) peer-reviewed jour- nal in its respective sub-category https://scholar.google.com/citations?view_op=top_venues&hl=en&vq=eng_ computingsystems

[8] Oxford Bioinformatics is considered *the* top (highest ranked) peer-reviewed journal in its respective sub-category https://scholar.google.com/citations?view_op=top_venues&hl=en&vq=eng_bioinformatics

Instruction-Level Parallelism to Cluster Mass Spectrometry Data using Multicore Architectures", *Springer Journal of Network Modeling Analysis in Health Informatics and Bioinformatics*, Volume 3, No. 1, pages 1-19, April 2014 **(Impact Factor: 2.3)**

35. **Fahad Saeed**†, Jason Hoffert and Mark Knepper, "CAMS-RS: Clustering Algorithm for Large-Scale Mass Spectrometry Data using Restricted Search Space and Intelligent Sampling", *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol.11, no.1, pp.128- 141, Feb. 2014 **(Impact Factor: 1.4)**
36. Jason Hoffert, Trairak Pisitkun, **Fahad Saeed**, Justin Wilson, and Mark Knepper, "Global Analysis of the Effects of the V2 Receptor Antagonist Satavaptan on Protein Phosphorylation in Collecting Duct", *American Journal of Physiology (AJP)*, Vol. 306, No. 410-421 February 2014 **(Impact Factor: 7.6)**
37. **Fahad Saeed**†, Trairak Pisitkun, Jason D. Hoffert, Sara Rashidian, Guanghui Wang, Marjan Gucek, and Mark A. Knepper, "PhosSA: Fast and Accurate Phosphorylation Site Assignment Algorithm for Mass Spectrometry Data", *Proteome Science* Volume 11, Supplement 1, November 2013 **(Impact Factor: 2.4)**
38. Pablo C. Sandoval, Dane H. Slentz, Trairak Pisitkun, **Fahad Saeed**, Jason D. Hoffert and Mark A. Knepper, "Proteome-wide measurement of protein half-lives and translation rates in vasopressin-sensitive collecting duct cells", *Journal of the American Society of Nephrology (JASN)*, Volume 24, Issue 11, pages 1793-1805, March 2013 **(Impact Factor: 9.3)**
39. Boyang Zhao, Trairak Pisitkun, Jason D. Hoffert, Mark A. Knepper, and **Fahad Saeed**†, "CPhos: a program to calculate and visualize evolutionary conserved functional phosphorylation sites", *PROTEOMICS*, Vol. 12, No. 22, pages 3299-3303, October 2012 **(Impact Factor: 3.8)**
40. Steven Bolger, Patricia Gonzales Hurtado, Jason Hoffert, **Fahad Saeed**, Trairak Pisitkun, and Mark Knepper, "Quantitative Phosphoproteomics in Nuclei of Vasopressin-Sensitive Renal Collecting Duct Cells", *American Journal of Physiology (AJP)*, Vol. 303, No. 10, pages C1006-C1020, Sept. 2012 **(Impact Factor: 7.6)**
41. Jacqueline Douglass, Ruwan Gunaratne, Davis Bradford, **Fahad Saeed**, Jason D. Hoffert, Peter J. Steinbach, Mark A. Knepper, and Trairak Pisitkun, "Identifying Protein Kinase Target Preferences Using Mass Spectrometry", *American Journal of Physiology (AJP)*, Vol. 303, No. 7, pages C715-C727, June 2012 **(Impact Factor: 7.6)**
42. **Fahad Saeed**, Alan Perez-Rathke, Jaroslaw Gwarnicki, Tanya Berger-Wolf, Ashfaq Khokhar, "A high performance multiple sequence alignment system for pyrosequencing reads from multiple reference genomes", *Journal of Parallel and Distributed Computing (JPDC)*, Volume 72, Issue 1, pages 83-93, January 2012 **(Impact Factor: 1.2)**
43. Jason D. Hoffert, Trairak Pisitkun, **Fahad Saeed**, Jae Song and Mark A. Knepper, "Dynamics of the vasopressin V2 receptor signalling network revealed by quantitative phosphoproteomics", *Molecular & Cellular Proteomics (MCP)*[9], Vol. 11, Issue No. 2, Feb 2012 **(Impact Factor: 6.9)**
44. Trairak Pisitkun, Jason D. Hoffert, **Fahad Saeed** and Mark Knepper, "NHLBI-AbDesigner: An online tool for design of peptide-directed antibodies", *American Journal of Physiology (AJP)*, Vol. 302, pages C154-C164, Jan 2012 **(Impact Factor: 7.6)**
45. **Fahad Saeed**, Ashfaq Khokhar, "A domain decomposition strategy for alignment of multiple biological sequences on multiprocessor platforms", *Journal of Parallel and Distributed Computing (JPDC)* Vol 69, Issue 7, pages 666-677, July 2009 **(Impact Factor: 1.2)**

7.6. Peer-Reviewed Conference Proceedings

46. Natalia Valencia, **Fahad Saeed**, and Mohammad Ashiqur Rahman, "Robust Cuffless Blood Pressure Estimation via PPG-Conditioned Diffusion and Contrastive Transformer Fusion", Proceeding of IEEE International Conference on Digital Health (ICDH), 2026 (Acceptance Rate: 32.4%).
47. Gabriel Bianchin de Oliveira, and **Fahad Saeed†,** "TITAN-BBB: Predicting BBB Permeability using Multi-Modal Deep-Learning Models", *Proceedings of 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (IEEE EMBC)*, (Full Paper), Toronto, Canada, July, 2026
48. Hamed Najafi, Pratik Poudel, Kiavash Bahreini, Julio Ibarra, **Fahad Saeed**, Yuepeng Li, Jayantha Obeysekera, Jason Liu, "RAPTOR: Reconfigurable Advanced Platform for Trans- disciplinary Open

[9] Molecular & Cellular Proteomics (MCP) is considered *the* top (highest ranked) peer-reviewed journal in its respective sub-category https://scholar.google.com/citations?view_op=top_venues&hl=en&vq=bio_proteomicspeptides

Research", Proceedings of *15th Workshop on AI and Scientific Computing at Scale using Flexible Computing Infrastructures (FlexScience)*, Notre Dame, IN, USA, July 20, 2025

49. Zichong Wang, Fang Liu, Shimei Pan, Jun Liu, **Fahad Saeed**, Meikang Qiu and Wenbin Zhang, "fairGNN-WOD: Fair Learning Without Complete Demographics", Proceedings of the *34th International Joint Conference on Artificial Intelligence (IJCAI)*, Montreal, Canada, 2025
50. Paras Parani, Muhammad Umair and **Fahad Saeed**†, "Utilizing Pretrained Vision Transformers and Large Language Models for Epileptic Seizure Prediction", Proceedings of *IEEE 8th International Conference on Data Science and Machine Learning Applications (CDMA)*, pp. 132-137, Riyadh, KSA, March 2025
51. Paras Parani, Muhammad Umair and **Fahad Saeed**†, "Lightweight Transformer exhibits comparable performance to LLMs for Seizure Prediction: A case for light-weight models for EEG data", 5th International Workshop on High Performance Computing, Big Data Analytics and Integration for Multi-Omics Biomedical Data (HPC-BOD), Proceedings of *IEEE Big Data Conference*, Washington DC USA, Dec. 15-18, 2024
52. Muhammad Umair and **Fahad Saeed**†, "Robustness of ML-Based Seizure Prediction Using Noisy EEG Data From Limited Channels", Proceedings of *20th Annual International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)*, Abudhabi, UAE, April 2024
53. Maryam Akhavan Aghdam, Serdar Bozdag, **Fahad Saeed**†, "PVTAD: Alzheimer's disease diagnosis using pyramid vision transformer applied to white matter of T-1 weighted structural MRI data", Proceedings of *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, May 27-30, 2024
54. Abhishek Bhattarai, Muhammad Umair, and **Fahad Saeed**†, "Communication Evaluation of a Wireless 4-Channel Wearable EEG for BCI and Health Applications", *IEEE SoutheastCon*, Atlanta Georgia USA, March 2024 (short paper)
55. Fahad Almuqhim and **Fahad Saeed**†, "ASD-GResTM: Deep Learning Framework for ASD classification using Gramian Angular Field", 7th International Workshop on Deep Learning in Bioinformatics, Biomedicine, and Healthcare Informatics (DLB2H), Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 2837-2843, Dec 5-8, 2023 (19.5% acceptance rate for long papers)
56. Muhammad Umair and **Fahad Saeed**†, "Energy Efficient AI/ML based Continuous Monitoring at the Edge: ECG and EEG Case Study", 14th International Workshop on High Performance Bioinformatics and Biomedicine (HiBB), Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 3313-3320, Dec 5-8, 2023 (19.5% acceptance rate for long papers)
57. Tianren Yang, Mai Al-Duailij, Serdar Bozdag, and **Fahad Saeed**†, "Classification of Autism Spectrum Disorder using Graph Convolutional Network and Graphlet Counting", 2nd International Workshop on Multi-Modal Medical Data Analysis, Proceedings of *IEEE International Conference on Big Data (IEEE BigData)*, Osaka Japan, pp. 3131-3138, Dec 17-20 2022. (19.2% acceptance rate)
58. Muhammad Umair and **Fahad Saeed**†, "SPERTL: Epileptic Seizure Prediction using EEG with ResNets and Transfer Learning", Proceedings of *IEEE International Conference on Biomedical and Health Informatics (BHI)*, Ioannina Greece, pp. 1-5, 27-30 September 2022. (30.2% acceptance rate)
59. Oswaldo Artiles and **Fahad Saeed**†, "A Multi-Factorial Assessment of Functional Human Autistic Spectrum Brain Network Analysis", International workshop on Reproducibility and Robustness in Biological Data Analysis (RROBIN), Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 3526-3531, December 9-12, 2021 (19.6% acceptance rate for regular papers)
60. Khandaker Mamun Ahmed, Taban Eslami, **Fahad Saeed**†, and M. Hadi Amini, "DeepCOVID- Net: Deep Convolutional Neural Network for COVID-19 Detection from Chest Radiographic Images", International workshop on Machine Learning for Biological and Medical Image Big Data, Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 1703-1710, December 9-12, 2021 (19.6% acceptance rate for regular papers)
61. Muhammad Usman Tariq, Dennys Leyva, Francisco Fernandez Lima, and **Fahad Saeed**†, "Graph Theoretic Approach for the Analysis of Comprehensive Mass-Spectrometry (MS/MS) Data of

Dissolved Organic Matter", International Workshop on Biological Network Analysis and Integrative Graph-Based Approaches(IWBNA), Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 3742-3746, December 9-12, 2021 (19.6% acceptance rate for regular papers)

62. Umair Muhammad, and **Fahad Saeed**†, "Simulation Testbed for Evaluating Distributed Query- ing and Searching of Mass Spectrometry Big Data in a Network-based Infrastructure", *Pro- ceedings of 7th IEEE International Conference on Big Data Service and Applications (IEEE BIGDATASERVICE)*, pp. 137-142, August 2021.
63. Umair Muhammad, and **Fahad Saeed**†, "Search Feasibility in Distributed MS-Proteomics Big Data", Workshop on High Performance Computing, Big Data Analytics and Integration for Multi-Omics Biomedical Data (HPC-BOD), *Proceedings of 12th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, article 85, August 2021.
64. Sumesh Kumar, and **Fahad Saeed**†, "Real-time peptide identification from high-throughput Mass-spectrometry data", Workshop on High Performance Computing, Big Data Analytics and Integration for Multi-Omics Biomedical Data (HPC-BOD), *Proceedings of 12th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, article 86, August 2021.
65. Oswaldo Artiles, and **Fahad Saeed**†, "TurboBC: A Memory Efficient and Scalable GPU Based Betweenness Centrality Algorithm in the Language of Linear Algebra", International Work- shop on Deployment and Use of Accelerators (DUAC), *50th International Conference on Parallel Processing (ICPP)*, Chicago IL, pp. 1-10, August 2021 (26.7% acceptance rate)
66. Sumesh Kumar, and **Fahad Saeed**†, "Communication-avoiding micro-architecture to compute Xcorr scores for peptide identification", *Proceedings of 31st International Conference on Field-Programmable Logic and Applications (FPL)*, pp. 99-103, May 2021 (22% acceptance rate)
67. Oswaldo Artiles, and **Fahad Saeed**†, "TurboBFS: GPU Based Breadth-First Search (BFS) Algorithms in the Language of Linear Algebra", 11th IEEE Workshop Parallel/ Distributed Combinatorics and Optimization (PDCO 2021), *IEEE International Parallel and Distributed Processing Symposium Workshops (IEEE IPDPS)*, pp. 520-528, May 2021 (22.7% acceptance rate)
68. Oswaldo Artiles, and **Fahad Saeed**†, "GPU-SFFT: A GPU based parallel algorithm for computing the Sparse Fast Fourier Transform (SFFT) of k-sparse signals", Workshop on Performance Engineering with Advances in Software and Hardware for Big Data Sciences (PEASH), Proceedings of *IEEE Conference on Big Data (IEEE BigData 2019)*, Los Angeles, CA, USA, pp. 3303-3311, Dec. 09-12, 2019. (18.7% acceptance rate)
69. Muhammad Haseeb, and **Fahad Saeed**†, "Efficient Shared Peak Counting in Database Pep- tide Search Using Compact Data Structure for Fragment-Ion Index", Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM)*, pp. 275-278, San Diego, CA, Nov 2019 (Acceptance Rate: 100/543=18%)
70. Mohammed Aledhari, Shelby Joji, Mohamed Hefeida, and **Fahad Saeed**†, "Optimized CNN-based Diagnosis System to Detect the Pneumonia from Chest Radiographs", Workshop on Computational Aspects for Clinical Diagnostics and Decision Making in Healthcare using Biomedical Signal and Image, Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM)*, pp. 2405-2412, San Diego, CA, Nov 2019
71. Taban Eslami, and **Fahad Saeed**†, "Auto-ASD-Network: A technique based on Deep Learning and Support Vector Machines for diagnosing autism spectrum disorder using fMRI data", Workshop on Machine Learning Models for Multi-omics Data Integration, in Proceedings of *10th ACM Conference on Bioinformatics, Computational Biology (ACM BCB)*, Niagara Falls, New York, pp. 646-651, September 7-10, 2019. (26.8% acceptance rate)
72. Muhammad Haseeb, Fatima Afzali, and **Fahad Saeed**†, "LBE: A Computational Load Balancing Algorithm for Speeding up Parallel Peptide Search in Mass-Spectrometry based Proteomics", Workshop of 18th IEEE International Workshop on High Performance Computational Biology (HICOMB), Proceedings of *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPS)*, Rio de Janeiro, Brazil, pp. 191-198, May 20, 2019
73. Taban Eslami, and **Fahad Saeed**†, "GPU-DFC: A GPU-based parallel algorithm for computing dynamic-functional connectivity of big fMRI data", Proceedings of *IEEE International Conference*

On Big Data Service And Applications (IEEE Big Data Service 2019), San Francisco East Bay, California, USA, pp. 114-121, April 4 - 9, 2019

74. Usman Tariq, and **Fahad Saeed**†, “Parallel Sampling-Pipeline for Indefinite Stream of Heterogeneous Graphs using OpenCL for FPGAs”, Workshop on Energy-Efficient Big Data Analytics, Proceedings of *IEEE International Conference on Big Data (IEEE BigData)*, pp. 4752-4761, Seattle, WA Dec 10-13, 2018 (19.7% acceptance rate)
75. Mohammed Aledhari, Marianne Di Pierro, and **Fahad Saeed**†, “A Fourier-Based Data Minimization Algorithm for Fast and Secure Transfer of Big Genomic Datasets”, Proceedings of *IEEE International Congress on Big Data (BigData Congress)*, pp. 128-134, San Francisco CA USA, pp. 128-134, July 2-7, 2018 (19.8% acceptance rate)
76. Taban Eslami, and **Fahad Saeed**†, “Similarity based classification of ADHD using Singular Value Decomposition”, Proceedings of *ACM International Conference on Computing Fron- tiers (ACM-CF)*, Ischia, Italy, pp. 19-25, May 2018 (33% acceptance rate)
77. Usman Tariq, Umer Cheema and **Fahad Saeed**†, “Power-Efficient and Highly Scalable Parallel Graph Sampling using FPGAs”, Proceedings of *International Conference on Reconfigurable Computing and FPGAs (ReConFig 2017)*, pp. 1-6, Cancun, Mexico, December 4-6, 2017
78. Sandino N. Vargas-Pérez and **Fahad Saeed**†, “Scalable Data Structure to Compress Next-Generation Sequencing Files and its Application to Compressive Genomics”, Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2017)*, pp. 1923- 1928, Kansas City, MO, USA, Nov 13-16, 2017 (18.6% acceptance rate)
79. Muaaz Gul Awan and **Fahad Saeed**†, “An Out-of-Core GPU based dimensionality reduction algorithm for Big Mass Spectrometry Data and its application in bottom-up Proteomics”, Proceedings of *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)*, pp. 550–555, Boston MA, August 20-23, 2017 (29% acceptance rate)
80. Taban Eslami, Muaaz Gul Awan and **Fahad Saeed**†, “GPU-PCC: A GPU based technique to compute pairwise Pearson’s Correlation Coefficients for big fMRI data”, Proceedings of *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)*, pp. 723–728, Boston MA, August 20-23, 2017 (29% acceptance rate)
81. Mohammed Aledhari, Ali Marhoon, Ali Al-Qaabi, and **Fahad Saeed**†, “A New Cryptography Algorithm to Protect Cloud-based Healthcare Services”, Proceedings of *IEEE/ACM Confer- ence on Connected Health: Applications, Systems and Engineering Technologies (IEEE/ACM CHASE-SEARCH)*, pp. 37–43, Philadelphia PA, July 2017
82. Muaaz Gul Awan and **Fahad Saeed**†, “GPU-ArraySort: A parallel, in-place algorithm for sorting large number of arrays”, Workshop on High Performance Computing for Big Data, Proceedings of *International Conference on Parallel Processing (ICPP-2016)*, pp. 78-87, Philadelphia PA, August 16-19, 2016. (21.1% acceptance rate)
83. Majdi Maabreh, Ajay Gupta and **Fahad Saeed**†, “A Parallel Peptide Indexer and Decoy Generator for Crux Tide using OpenMP”, Workshop on High Performance Computing Systems for Biomedical, Bioinformatics and Life Sciences, Proceedings of *International Conference on High Performance Computing & Simulation (HPCS 2016)*, Innsbruck, Austria, pp. 411-418, July 2016 **(Nominated for Best Paper Award)**
84. Mohamed S Hefeida and **Fahad Saeed**†, “Data Aware Communication for Energy Harvest- ing Senor Networks”, Proceedings of *International Conference on Wired & Wireless Internet Communications (WWIC 2016)*, Thessaloniki, Greece, pp. 121-132, May 2016 (27 papers accepted out of 55 papers submitted: 49% acceptance rate)
85. Mohammed Aledhari, Mohamed S Hefeida and **Fahad Saeed**†, “A Variable-Length Network Encoding Protocol for Big Genomic Data”, Proceedings of *International Conference on Wired & Wireless Internet Communications (WWIC 2016)*, Thessaloniki, Greece, pp. 212-224, May 2016 (27 papers accepted out of 55 papers submitted: 49% acceptance rate)
86. **Fahad Saeed**†, “Big Data Proteogenomics and High-Performance Computing: Challenges and Opportunities”, Proceedings of *Symposium on Signal and Information Processing for Software-Defined Ecosystems, and Green Computing, IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, Orlando Florida, Dec 2015
87. Sandino N. Vargas-Pérez and **Fahad Saeed**†, “A Parallel Algorithm for Compression of Big Next-Generation Sequencing (NGS) Datasets”, Proceedings of *Parallel and Distributed Processing with Applications (IEEE ISPA-15)* Vol.3. pp. 196-201, Helsinki Finland, August 2015

88. Mohammed Aledhari and **Fahad Saeed**†, "Design and Implementation Network Transfer Protocol for Big Genomic Data", Proceedings of *IEEE Big Data Congress*, pp.281-288, New York City, USA, June 2015 (18% acceptance rate)
89. Muaaz Gul Awan and **Fahad Saeed**†, "On the sampling of Big Mass Spectrometry Data", Proceedings of *ISCA International Conference on Bioinformatics and Computational Biology (BICoB)*, Honolulu, Hawaii, USA, pp. 143-148, March 2015
90. **Fahad Saeed**†, Jason Hoffert, and Mark Knepper, "A High Performance Algorithm for Clustering of Large-Scale Protein Mass Spectrometry Data using Multi-Core Architectures", Proceedings of *IEEE/ACM International Symposium on Network Enabled Health Informatics, Biomedicine and Bioinformatics (HI-BI-BI)*, pp. 923 - 930, August 2013 (25% acceptance rate for full papers)
91. **Fahad Saeed**†, Trairak Pisitkun, Jason Hoffert, Guanghui Wang, Marjan Gucek, and Mark Knepper, "An Efficient Dynamic Programming Algorithm for Phosphorylation Site Assign- ment of Large-Scale Mass Spectrometry Data", International Workshop on Computational Proteomics, Proceedings of *Proceedings of IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Philadelphia USA, pp. 618-625, Oct 2012 (20% acceptance rate)
92. **Fahad Saeed**†, Trairak Pisitkun, Jason Hoffert, and Mark A. Knepper, "High Performance Phosphorylation Site Assignment Algorithm for Mass Spectrometry Data using Multicore Systems", proceedings of *ACM Conference on Bioinformatics, Computational Biology and Biomedicine (ACM-BCB)*, pp. 667-672, Orlando Florida USA, Oct 2012. (33 papers accepted out of 159 papers submitted: 21% acceptance rate)
93. **Fahad Saeed**†, Trairak Pisitkun, Mark A. Knepper, and Jason D. Hoffert, "An Efficient Algorithm for Clustering of Large-Scale Mass Spectrometry Data", Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 1-4, Philadelphia USA, Oct 2012. (62 short paper accepted out of 299 papers submitted: 20.7 % acceptance rate)
94. **Fahad Saeed**† and Ashfaq Khokhar, "Parallel algorithm for center-star sequence alignments with applications to short reads" Proceedings of *ISCA International Conference on Bioinformatics and Computational Biology (BICoB)*, Las Vegas Nevada, USA, March 2012
95. **Fahad Saeed**†, Trairak Pisitkun, Mark A. Knepper, and Jason D. Hoffert, "Mining Temporal Patterns from iTRAQ Mass Spectrometry (LC-MS/MS) Data" Proceedings of *ISCA Bioinformatics and Computational Biology Conference (BICoB)* pp 152-159, New Orleans USA, March 2011 .
96. **Fahad Saeed**, Ashfaq Khokhar, Osvaldo Zagordi and Niko Beerenwinkel, "Multiple Sequence Alignment System for Pyrosequencing Reads", Proceedings of International Conference on Bioinformatics and Computational Biology (BICoB), in *Lecture Notes in Computer Science (LNCS)*, Volume 5462/2009, pp 362-375, 2009.
97. **Fahad Saeed** and Ashfaq Khokhar, "Sample-Align-D: A High-Performance Multiple Sequence Alignment System using Phylogenetic Sampling and Domain Decomposition", Proceedings of *IEEE International Symposium on Parallel and Distributed Processing (IPDPS)*, Monday, April 2008.

7.7. Peer-Reviewed Abstracts and Posters

98. Chloe L. Hampson; Julio A. Peraza; Lauren M. Guerrero; Katherine L. Bottenhorn; Michael C. Riedel; Fahad Almuqhim; **Fahad Saeed**; Donisha D. Smith; Katherine M. Schmarder; Katharine E. Crooks; Rosario Pintos Lobo; Matthew T. Sutherland; Erica D. Musser; Yael Dai; Rumi Agarwal; Angela R. Laird, "Habenula Alterations in Resting State Functional Connectivity among Autistic Individuals", *Organization for Human Brain Mapping (OHBM)*, Brisbane, Australia, June 24-28, 2025
99. Pablo Batista Oliveira, Lilian Valadares Tose, Kevin Jeanneditfouque, Bilal Shabbir, David Pakilgour, **Fahad Saeed**, Francisco Alberto Fernandez Lima, "Structural Characterization of DOM using TIMS-FT ICR MS/MS and Machine Learning", *ASMS Conference on Mass Spectrometry and Allied Topics*, Baltimore MD, June 2-6, 2025
100. Bilal Shabbir; Usman Tariq; Syed Talha Khalid; Francisco Fernandez-Lima; **Fahad Saeed**†, "Beyond Mass-Only Filtering: A Multitask Deep Learning Framework for Predicting Peptide Properties from Mass", *ASMS Conference on Mass Spectrometry and Allied Topics*, Baltimore MD, June 2-6, 2025
101. Tahmina Rahman; **Fahad Saeed**, Serdar Bozdag, "Alzheimer's disease-associated gene ranking using PhenoGeneRanker", Alzheimer's & Dement. Journal of Alzheimer's Association

(20(Suppl 2):e089944) :1-5 (2025). Alzheimer & Dementia

102. Yashu Vashishath; Sarah Beaver; **Fahad Saeed**; Serdar Bozdag, “Predicting Individual’s Cognitive Performance Through Multi-Omics Blood Data Using Hierarchical Input Neural Network - HINN”, Alzheimer's & Dement. Journal of Alzheimer's Association (20(Suppl 8):e094598) :1-5 (2025). Alzheimer & Dementia
103. Maryam Akhavan Aghdam; Serdar Bozdag; **Fahad Saeed**, “Alzheimer’s disease diagnosis using gray matter of T1-weighted sMRI data and vision transformer”, Alzheimer's & Dement. Journal of Alzheimer's Association (20(Suppl 2):e089944) :1-5 (2025), Alzheimer’s & Dementia
104. Mohammad Al Olaimat, Serdar Bozdag and **Fahad Saeed**, “TA-RNN: an Attention-based Time-aware Recurrent Neural Network Architecture to Predict Progression of Alzheimer’s Disease”, Alzheimer’s Dement. Vol. 20, Suppl. 1, e089010, 2025
105. Hampson, C. L., Peraza, J. A., Guerrero, L. M., Bottenhorn, K. L., Riedel, M. C., Almuqhim, **Fahad Saeed**, Smith, D. D., Schmarder, K. M., Crooks, K. E., Viera Perez, P. M., Musser, E. D., Dai, Y. G., Agarwal, R., Sutherland, M. T., & Laird, A. R., “Habenula alterations in resting state functional connectivity in autism spectrum disorder”, Poster presented at the *University of Miami’s 34th Annual Neuroscience Research Day*, Miami, FL 2024.
106. Yashu Vashishath, Sarah Beaver, **Fahad Saeed** and Serdar Bozdag, “HINN: A Novel Neural Network Architecture to Integrate Multi-Omics Data based on their Biological Relationships”, *ISCB’s Intelligent Systems for Molecular Biology (ISMB)*, Montreal, Canada, July 15 2024
107. Maryam Akhavan Aghdam, Serdar Bozdag, **Fahad Saeed**†, “Alzheimer’s disease diagnosis using gray matter of T1-weighted sMRI data and vision transformer”, *Alzheimer’s Association International Conference (AAIC)*, Philadelphia PA USA, July 29, 2024.
108. Pablo Batista Oliveira, Dennys Leyva Bombuse, Lilian Valadares Tose, Muhammad Usman Tariq, **Fahad Saeed**, Francisco Alberto Fernandez Lima, “High Resolution Ion Mobility and Ultra-High resolution mass spectrometry for DOM chemical Formula based Structural Assignment”, *ASMS Conference on Mass Spectrometry and Allied Topics*, Anaheim, CA, June 2-6, 2024
109. Andrew Forero, Miguel Santos Fernandez, Pablo Batista Oliveira, Kevin Jeanneditfouque, Cuellen Green, Brian Clowers, **Fahad Saeed**, Francisco Alberto Fernandez Lima, “Integration of tandem high resolution ion mobility and mass spectrometry for complex mixture analysis”, *ASMS Conference on Mass Spectrometry and Allied Topics*, Anaheim, CA, June 2-6, 2024
110. Fahad Almuqhim, and **Fahad Saeed**† , “Effects of multisite data and harmonization of fMRI data and its effects on machine-learning models”, *USF Artificial Intelligence + X Symposium*, Tampa, FL September 29, 2023
111. Umair Muhammad, and **Fahad Saeed**†, “Towards real-world ML/AI models for epileptic seizure prediction from wearable EEG”, *USF Artificial Intelligence + X Symposium*, Tampa, FL September 29, 2023
112. Fahad Almuqhim, and **Fahad Saeed**†, “Explore the effects of multi-site data acquisition on functional magnetic resonance imaging (fMRI) data”, *Resting State Brain Connectivity Conference (RSBC)*, Dallas, Texas, September 18, 2023
113. Robert Loredo, and **Fahad Saeed**†, “Quantum-Centric Supercomputing Strategies for Neuroscience problems: Challenges and Progress”, Quantum Computing Algorithms, Systems, and Applications (Q-CASA) Workshop, *International Parallel and Distributed Processing Symposium (IPDPS)*, May 15-19, 2023, St. Petersburg, FL **(Invited Abstract)**
114. Muhammad Haseeb, and **Fahad Saeed**†,“GPU-Acceleration of the Distributed-Memory Database Peptide Search on Supercomputers”, *ASMS Conference on Mass Spectrometry and Allied Topics*, Minneapolis, MN May 2022
115. Usman Tariq, and **Fahad Saeed**†, “Improving MS/MS-based Peptide Database Search using Deep Learning”, *ASMS Conference on Mass Spectrometry and Allied Topics*, Minneapolis, MN May 2022
116. Paulina Acousta, Usman Tariq, **Fahad Saeed**†, “Deep-Learning Network for Classifying Diseases using Mass Spectrometry Data”, *Undergraduate Research Conference FIU*, March 2022 Miami FL
117. Usman Tariq, Francisco Alberto Fernandez Lima, and **Fahad Saeed**† , “Towards a Computational Workflow for the Analysis of DOM Fragmentation Data”, *ASMS Conference on Mass Spectrometry and Allied Topics*, Houston, Texas, May 31 - June 4, 2020

118. Hyun Jun Jung, Bronte Wen, Lihe Chen, Fahad Saeed and Mark A. Knepper, “NGS-Integrator: A Tool for Combining Information from Multiple Genome-Wide NGS Data Tracks Using Minimum Bayes Factors”, *The FASEB Journal*, Vol. 33, No. 1 supplement April 2019
119. Jae Wook Lee, Chung-Lin Chou, **Fahad Saeed**, and Mark A. Knepper, “RNA-seq of Microdissected Renal Tubules Identifies Segment-Specific Transcription Factors”, poster presentation at *American Society of Nephrology (ASN)*, August 2013.
120. Jae Wook Lee, Chung-Lin Chou, **Fahad Saeed**, and Mark A. Knepper, “RNA-seq Identification of Transcriptome of Native DCT1 Cells”, poster presentation at *American Society of Nephrology (ASN)*, August 2013.
121. Jae W. Lee, **Fahad Saeed**, Chung Lin Chou and Mark A. Knepper, “RNA-Seq Mapping Of G Protein-Coupled Receptor Expression along the Nephron and Collecting Duct”, poster presentation at *American Society of Nephrology (ASN)*, August 2013.
122. Steven J Bolger, Patricia A Gonzales, Jason D Hoffert, **Fahad Saeed**, Trairak Pisitkun and Mark A Knepper, “Quantitative phosphoproteomics implicates clusters of proteins involved in cell-cell adhesion and transcriptional regulation in the vasopressin signaling network”, meet- ing abstracts *Experimental Biology (EB)*, July 2013
123. Boyang Zhao, Trairak Pisitkun, Jason D. Hoffert, Mark A. Knepper, and **Fahad Saeed**†, “An Information Theory-Based Approach to Assess the Functional Significance of Phosphorylation Sites in Proteomes of Renal Tubule Epithelia”, poster at *International Society of Nephrology (ISN) Symposium*, Ann Arbor Michigan, USA, June 2012
124. **Fahad Saeed**†, J. Hoffert, P. Pisitkun, M. Knepper, “Mapping-based temporal pattern mining algorithm identifies unique clusters of phosphopeptides regulated by vasopressin in collecting duct”, meeting abstracts *Experimental Biology (EB)*, Washington DC, USA, April 2011.
125. J. Hoffert, T. Pisitkun, **Fahad Saeed**, J. Song, M. Knepper, “Large-scale iTRAQ-based quantification of phosphorylation changes during vasopressin signaling”, Featured Topic and abstract *Experimental Biology (EB)*, Washington DC USA, April 2011.
126. **Fahad Saeed** and Ashfaq Khokhar, “Parallel Algorithm for Center Star Sequence and Alignments with applications to short reads”, International Conference On *Bioinformatics and Computational Biology (ACM-BCB)* in August 2010.

7.8. Talks and Presentations

1. “Capturing Proteomic and Neuro Architecture Complexity using Advance Machine Learning Modelling”, *School of Computing, Faculty of Computing, Engineering and the Built Environment, Ulster University at Belfast Campus, Northern Ireland, UK*, Feb 11th, 2026 (**Invited Distinguished Research Seminar)**
2. “Scaling and Modeling Computational (AI/ML) Methods for Population Systems”, *Population Health Initiative (PHI), Florida International University (FIU)*, Miami FL, Sept 25th, 2025 (**Panelist and Invited Talk**)
3. “Computational Journey: Scaling and Modeling Systems - (Biological and Health) – Complexity”, *Multi-Interprofessional Center for Health Informatics (MICHI), University of Texas at Arlington (UTA)*, Arlington TX, USA, Oct 3rd, 2024
4. “Computational Journey: Scaling and Modeling Systems - (Biological and Health) – Complexity”, *Knight Foundation School of Computing and Information Sciences)*, College of Engineering and Computing, Florida International University (FIU), Miami, FL USA Sept 27, 2024
5. “Personalized medicine: How you (computer scientist) can make a difference”, *Florida IT Graduation Attainment Pathways (Flit-GAP)*, College of Engineering and Computing, Florida International University (FIU), Miami, FL USA Sept 6th, 2024.
6. “Scalable machine-learning models for healthcare”, *Biomedical Engineering Department*, College of Engineering, University of Miami, Coral Gables, FL USA May 13th, 2024.
7. “Graph-DOM: Graph Theoretic Approach for Analyzing Dissolved Organic Matter”, *FIUMASS Winter DOM school*, conducted in collaboration with Dr. Francisco Lima, Florida International University (FIU) Miami FL, Feb 28th, 2024
8. “Quantum-Centric Supercomputing Strategies for Neuroscience problems: Challenges and Progress”, *Quantum Computing Algorithms, Systems, and Applications (Q-CASA), IPDPS*, St. Petersburg, FL USA, May 15th, 2023 **(Invited Talk)**
9. “Importance of high-performance computing for big data computational biology”, Opening remarks

as program co-chair at the *22nd IEEE International Workshop on High Performance Computational Biology (HiCOMB), IPDPS*, St. Petersburg, FL USA, May 15th, 2023

10. "Solving Grand Challenges in Proteomics and Neuroscience with High Performance Data Analytics and Machine-Learning models: Challenges, and Progress", *Electrical and Computer Engineering, University of Miami*, Coral Gables, FL USA April 27th, 2023.
11. "Fi-Sci Analogies to narrate Science and complex ideas", with Marie Alexandra Cornelius, Nicola Gavioli, Saby Moulik, Zion Michael, Jeanette Smith, Tigran Abrahamyan, and **Fahad Saeed**, *International Conference on Narrative*, Dallas TX, USA, March 4th, 2023 **(Keynote Panel)**
12. "Machine Learning: Applications and Opportunities in Biomedical Science", *Dept. of Computing Sciences, University of Houston - Clear Lake (UHCL)*, Houston, TX, USA, Nov 29th 2022 **(Invited Talk)**
13. "Data Analytics and Machine-learning models to solve problems in Proteomics and Neuro-science", *Electrical and Computer Engineering, University of Miami*, Coral Gables, FL USA, Nov 15th 2022 **(Invited Talk)**
14. "Solving Grand Challenges in Proteomics and Neuroscience with High Performance Data Analytics and Machine-Learning models: Progress and Lessons Learned", *14th International Conference on Bioinformatics and Computational Biology (BICOB)*, March 22nd, 2022 **(Keynote Speaker)** https://sceweb.sce.uhcl.edu/bicob22/invited_speakers.html
15. "Machine-learning and HPC Models for Big Data Proteomics", *15th IEEE International Conference on Open Source Systems & Technologies (ICOSST 2021)*, Dec 16th 2021 **(Distiguished invited talk)** https://icosst.kics.edu.pk/2021/invited-speakers/
16. "Machine-learning Models for Big Data Omics: Challenges, Opportunities and Progress", *Machine Learning Focus Group KFSCIS*, Florida International University (FIU) MMC cam- pus, Miami FL USA, April 20th, 2021 **(Invited Talk)**
17. "Machine-learning Models for Big Data Omics: Challenges, Opportunities and Progress", *Biomolecular Sciences Institute*, Florida International University (FIU) MMC campus, Miami FL USA, Feb 26th, 2021 **(Invited Talk)**
18. "Importance of high-performance computing in the context of multiomics data analysis, machine learning models", *ACM-BCB HPC-BOD Workshop Introductory chair remarks*, Virtual, Sept 21, 2020
19. "HPC Techniques for Big Mass Spectrometry Proteomics Data using GPU's", *National Energy Research Scientific Computing Center (NERSC), Lawrence Berkeley National Laboratory*, Berkeley, CA, USA June 30th, 2020 **(Invited Talk)**
20. "Compact Indexing for Fragment-Ion Efficient Shared-Peak Counting from Mass Spectrometry Proteomics Data", *IEEE BIBM*, San Diego CA, USA Nov 2019
21. "Extreme Scale Omics and Next Generation of Computational Infrastructure", *IEEE BIBM*, San Diego CA, USA Nov 2019 (Invited Talk) [declined due to scheduling conflict]
22. "Supercomputing and Machine-Learning for Big Omics data", Research Day at KFSCIS, *Florida International University (FIU)*, MMC campus, Miami, Florida USA, Oct 2019
23. "Challenges and Entrepreneurial Opportunities in Big Data Omics", at the FIU KFSCIS Industry Advisory Board Meeting, *Florida International University (FIU)*, Miami, Florida USA, April 2019
24. "Extreme Scale Omics and Next Generation of Computational Infrastructure", at the School of Computing and Information Sciences, *Florida International University (FIU)*, Miami, Florida USA, August 2018
25. "Data-Reduction Algorithms for Big Data Proteomics: Challenges, Opportunities and Progress", at the Department of Mathematics and Statistics, *Western Michigan University (WMU)*, Kalamazoo, MI USA, April 2018 **(Invited)**
26. "Big Data Proteogenomics and High-Performance Computing: Challenges, Progress and Opportunities", at the School of Computing and Information Sciences, *Florida International University (FIU)*, Miami, Florida, USA Jan 2018
27. "Reductive and HPC Analytics for Big Mass Spectrometry Data and its applications to Proteomics", *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, Boston MA, USA August 2017
28. "Big Data Proteogenomics and High-Performance Computing: Challenges and Opportunities", *Symposium on Signal and Information Processing for Software-Defined Ecosystems, and Green Computing, (IEEE GlobalSIP)*, Orlando Florida USA, Dec 2015

29. “High Performance Computing and Big Data Computational Biology”, introductory remarks at the inaugural *IEEE Workshop of High-Performance Computing and Big Computational Biology (HPC-BCB), IEEE BIBM*, Washington DC, USA, Nov 2015
30. “Big Data Proteomics: High Performance Computing Methods & Applications”, at School of Science & Engineering, *Lahore University of Management Science (LUMS)*, Lahore Pakistan, January 2015 **(Invited Seminar)**
31. “Solving Big Data Problems in Computational Biology using High Performance Architectures and Algorithms”, at Department of Electrical Engineering and Computer Systems, *University of Cincinnati*, Cincinnati Ohio, USA November 2013
32. “Big Data Problems in high-throughput Genomics and Proteomics”, at *Cincinnati Children’s Hospital Medical Center*, Cincinnati Ohio, USA, November 2013
33. “Solving Big Data Problems in Computational Biology using High Performance Architectures and Algorithms”, at Department of Computer Science, *Western Michigan University*, Kalamazoo MI, USA, October 2013
34. “Solving Big Data Problems in Computational Biology using High Performance Architectures and Algorithms”, at Department of Electrical and Computer Engineering, *Western Michigan University*, Kalamazoo MI, USA, August 2013
35. “Solving Big Data Problems in Computational Biology using High Performance Architectures and Algorithms”, at Innovation Center for Biomedical Informatics, *Georgetown University Medical Center*, Washington DC, USA, August 2013
36. “Clustering and Consensus of Large-Scale Mass Spectrometry Data”, Work in Progress Series, Systems Biology Interest Group, National Heart, Lung and Blood Institute (NHLBI), *National Institutes of Health (NIH)*, Bethesda MD, USA May 2013.
37. “High Performance Algorithm Engineering for Large-Scale Next Generation Genomics (NGS) Data”, at the Computer Science Department, *University of Massachusetts Boston*, Boston MA, USA, March 2013.
38. “High Performance Algorithm Engineering for Large-Scale Next Generation Genomics (NGS) Data”, at Electrical, Computer and Systems Engineering Department, *Rensselaer Polytechnic Institute (RPI)*, Albany NY, USA, March 2013.
39. “High Performance Algorithm Engineering for Large-Scale Next Generation Genomics (NGS) Data”, at Electrical and Computer Engineering Department, *George Washington University*, Washington DC, USA Feb 2013.
40. “High Performance Algorithm Engineering for Large-Scale Next Generation Sequencing (NGS) Data”, at Center for Genome Research and Biocomputing (CGBC), *Oregon State University*, Corvallis Oregon, USA, Oct 2012.
41. “An Efficient Algorithm for Clustering of Large-Scale Mass Spectrometry Data”, *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Philadelphia Pennsylvania, USA, Oct 2012
42. “An Efficient Dynamic Programming Algorithm for Phosphorylation Site Assignment of Large-Scale Mass Spectrometry Data”, International workshop on computational proteomics, *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Philadelphia Pennsylvania, USA, Oct 2012
43. “HPC approaches to computational problems in high-throughput genomics and proteomics”, at Electrical and Computer Engineering Department, *State University of New york (SUNY) at Binghamton*, Binghamton NY, USA, Feb 2012
44. “A graph theoretic approach to spectral clustering of mass spectrometry data”, Work in Progress Series, Systems Biology Interest Group, National Heart, Lung and Blood Institute (NHLBI), *National Institutes of Health (NIH)*, Bethesda MD, USA, Jan 2012.
45. “Computational problems in large-scale protein phosphorylation studies using iTRAQ labeled LC MS/MS data”, *Bioinformatics and Computational Biology Conference (BICoB)*, New Orleans, Louisiana, USA, March 2011.
46. “Dynamic Programming Algorithm for Phosphorylation Site Assignment of Mass Spectrometry Data”, Work in Progress Series, Systems Biology Interest Group, National Heart, Lung and Blood Institute (NHLBI), *National Institutes of Health (NIH)*, Bethesda MD, USA, Feb 2011.
47. “Challenges in multiple alignment of huge number of pyrosequencing reads on multiprocessor platforms”, at *Institute of Genetic Medicine, Keck School of medicine, University of Southern*

California (USC), Los Angeles, CA USA, March 2010.

48. "Challenges in multiple alignment of huge number of pyrosequencing reads on multiprocessor platforms", at *school of medicine, University of Colorado at Denver (UCD)*, Denver, CO, USA Feb 2010.
49. "How to multiple align huge number of short reads", at *Critical Assessment of Massive Data Analysis (CAMDA)*, Chicago IL, USA, Oct 2009. **(Invited)**
50. "An Efficient Multiple Alignment system for Pyrosequencing Reads", at *Bioinformatics and Computational Biology Conference (BICoB)*, New Orleans, LA, USA, April 09.
51. "A Domain Decomposition strategy for Multiple Alignment of Biological Reads", at *Laboratory of Computational Population Biology, University of Illinois at Chicago*, Chicago, IL USA, Nov 08. **(Invited)**
52. "A Sampling based High Performance Multiple Alignment systems", *IEEE International Workshop on High Performance Computational Biology, IPDPS*, Miami Florida, USA, April 08.

7.9. Thesis

1. **Fahad Saeed**, "High performance computational biology algorithms", (Doctoral dissertation). Retrieved from ProQuest Dissertations and Theses (Accession Order No. AAT 3431281, ISBN: 9781124308067), 2010.
2. **Fahad Saeed**, Fahad A. Arshad and Salman Javed, "Design and Implementation of Linux Based Wireless Ad-Hoc Network based Compute Laboratory", (Undergraduate Final Report), Submitted as partial requirements for BSc. degree in Electrical Engineering University of Engineering and Technology Lahore, 2005.

7.10. Technical Reports

1. Oswaldo Artiles, **Fahad Saeed (Editor)**, "Statistical and Machine Learning Analysis of the Human Brain Functional Network in a Multi-Site Resting-State Functional MRI Database Framework", FIU School of Computing and Information Sciences Technical Report (39), 2023. (Weblink: https://digitalcommons.fiu.edu/cs_fac/30/)
2. Muhammad Haseeb**, Fahad Saeed (Editor),** "High Performance Computing Algorithms for Accelerating Peptide Identification from Mass-Spectrometry Data Using Heterogeneous Supercomputers, FIU School of Computing and Information Sciences Technical Report (39), 2023. (Weblink: https://digitalcommons.fiu.edu/cs_fac/29)
3. **Fahad Saeed**, Lukas Burger, Ashfaq Khokhar, and Mihaela Zavolan, "A graph-theoretic framework for efficient computation of HMM based motif finder", Technical Report, *University of Illinois at Chicago*, Jan 2010.
4. **Fahad Saeed**, "High Performance Graph Theoretic model for finding Regulatory Elements and motifs", Technical Report, Zavolan Group, *Swiss Institute of Bioinformatics (SIB), University of Basel Switzerland*, August 2009.
5. **Fahad Saeed**, "High Performance Multiple Alignment Systems for Pyrosequencing Reads of very large number", Technical Report, Beerenwinkel Group Computational Biology, *Department of Biosystems Science and Engineering, Eth Zurich* Switzerland, August 2008.
6. **Fahad Saeed** and Ashfaq Khokhar, "An overview of parallel programs for multiple sequence alignments and their limitations", technical report, *Parallel Algorithms and Multimedia System Laboratory, University of Illinois at Chicago*, May 2007.
7. **Fahad Saeed**, "Securing Suparco's Computer Networks and Linux Operating Systems", Internal research Report on *Suparco's Computer Networks Security* X11/04/2006-I, made official in April 2006.
8. **Fahad Saeed**, Fahad Ali Arshad and Salman Javed, "Design and Implementation of Linux based wireless ad-hoc network", *Linux Focus*, article 390, Dec 2005.

7.11. Software (Released under open-source license or provided as a web-service)

1. HiCOPS

Database peptide search algorithms deduce peptides from mass spectrometry data. There has been substantial effort in improving their computational efficiency to achieve larger and more complex systems biology studies. However, modern serial and high-performance computing

(HPC) algorithms exhibit suboptimal performance mainly due to their ineffective parallel designs (low resource utilization) and high overhead costs. We present an HPC frame- work, called HiCOPS, for efficient acceleration of the database peptide search algorithms on distributed-memory supercomputers. HiCOPS provides, on average, more than tenfold improvement in speed and superior parallel performance over several existing HPC database search software. We also formulate a mathematical model for performance analysis and optimization, and report near-optimal results for several key metrics including strong-scale efficiency, hardware utilization, load-balance, inter-process communication and I/O overheads. The core parallel design, techniques and optimizations presented in HiCOPS are search-algorithm-independent and can be extended to efficiently accelerate the existing and future algorithms and software. Our HPC framework can be downloaded from the following link: https://hicops.github.io/.

2. **SpeCollate**

We designed and developed a Deep Cross-Modal Similarity Network called SpeCollate, which overcomes these inaccuracies by learning the similarity function between experimental spectra and peptides directly from the labeled MS data. SpeCollate transforms spectra and peptides into a shared Euclidean subspace by learning fixed size embeddings for both. Our proposed deep-learning network trains on sextuplets of positive and negative examples coupled with our custom-designed SNAP-loss function. Online hardest negative mining is used to select the appropriate negative examples for optimal training performance. We use 4.8 million sextuplets obtained from the NIST and MassIVE peptide libraries to train the net- work and demonstrate that for closed search, SpeCollate is able to perform better than Crux and MSFragger in terms of the number of peptide-spectrum matches (PSMs) and unique peptides identified under 1% FDR for real-world data. SpeCollate also identifies a large number of peptides not reported by either Crux or MSFragger. To the best of our knowl- edge, our proposed SpeCollate is the first deep-learning network that can determine the cross-modal similarity between peptides and mass-spectra for MS-based proteomics. We believe SpeCollate is significant progress towards developing machine-learning solutions for MS-based omics data analysis. Our machine-learning model can be downloaded from the following link: https://deepspecs.github.io/.

3. **ASD-DiagNet**

We propose a framework called ASD-DiagNet for classifying subjects with Autism Spectrum Disorder (ASD) from healthy subjects by using only fMRI data. We designed and implemented a joint learning procedure using an autoencoder and a single layer perceptron which results in improved quality of extracted features and optimized parameters for the model. Further, we designed and implemented a data augmentation strategy, based on linear inter- polation on available feature vectors, that allows us to produce synthetic datasets needed for training of machine learning models. The proposed approach is evaluated on a public dataset provided by Autism Brain Imaging Data Exchange including 1035 subjects coming from 17 different brain imaging centers. Our machine learning model outperforms other state of the art methods from 13 imaging centers with an increase in classification accuracy up to 20% with maximum accuracy of 80%. The machine learning technique presented in this paper, in addition to yielding better quality, gives enormous advantages in terms of execution time (40 minutes vs. 6 hours on other methods). The tool can be downloaded from the following link: https://github.com/pcdslab/ASD-DiagNet

4. **MaSS-Simulator**

MaSS-Simulator is capable of simulating MS/MS spectra for LC-MS/MS based proteomics experiments. Our recently introduced MaSS-Simulator is capable of simulating highly accurate MS/MS spectra for LC-MS/MS based proteomics experiments. It provides a great degree of control over the simulation by providing multiple configurable parameters. MaSS-Simulator offers a platform to assess and mark the limitations of MS-proteomics algorithms by testing them against a curated set of data and a whole range of parameters. A complete evaluation report of an algorithm using all possible parametric verifications will provide a much deeper insight to the performance of a given algorithm. Such evaluation will solve the reproducibility issues that are frequently faced in proteomics algorithm development. Testing of tools using such curated data set for which parameters (e.g. peptide coverage, S/N ratio etc.) can be carefully explored will rigorously evaluate the proteomics algorithms. In general, one can vary (to a practical degree) the size of the peptide, the coverage, the S/N ratios of the spectra and addition of PTM's in the simulated spectra created for benchmarking. Further, such reproducible evaluation of proteomics algorithms will enable method developers to endorse algorithms with confidence and reliability. Such an approach will also be helpful for the users, who can evaluate their dataset and cross-reference its properties with the algorithm's evaluation report to conclude if the given algorithm will serve their purpose or if it will be able to achieve the required level of performance. The tool can be downloaded from the following link: https://github.com/pcdslab/MaSS-Simulator

5. **J-Eros**

J-Eros is a machine learning algorithm for computing similarity between two multivariate time series along with k-Nearest-Neighbor classifier, to classify healthy vs ADHD children using just fMRI data without using any other data or demographics. We applied this technique to the public data provided by the ADHD-200 Consortium competition and our results show that J-Eros is capable of discriminating healthy from ADHD children such that we outperformed other state of the art techniques. This machine learning algorithm is a major step towards diagnosing ADHD using quantitative methods and will be an essential part for diagnosing mental illnesses. The algorithm is developed in python and can be accessed at https://github.com/pcdslab/J-Eros

6. **PHYNGSC**

phyNGSC is a hybrid strategy between MPI and OpenMP to accelerate the compression of big FASTQ datasets by combining the best features of distributed and shared memory architectures to balance the load of work among processes, alleviate memory latency by exploit- ing locality and accelerate I/O by reducing excessive read/write operations and inter-node message exchange. The algorithm introduces a novel timestamp-based approach which allows concurrent writing of compressed data in a non-deterministic order and thereby allows us to exploit a high amount of parallelism. As a proof-of-concept, we implemented some methods developed for DSRC v1 to underline the compression portion of our hybrid parallel strategy, since it exhibits superior performance for sequential solutions. The parallel algorithm is developed using C/C++, MPI and OPENMP constructs and is available at https://github.com/pcdslab/PHYNGSC.

7. **GPU-ArraySort**

GPU-ArraySort is a highly scalable parallel algorithm for sorting large numbers of arrays using a GPU. Existing techniques focus on sorting a single large array and cannot be used for sorting large number of smaller arrays in an efficient manner. Such small numbers of large arrays are common in many big data applications in fields such as proteomics, genomics, connectomics, and astronomy. Our algorithm performs in-place operations and makes minimum use of any temporary run-time memory. Our results indicate that we can sort up to 2 million arrays having 1000 elements each, within a few seconds. We compare our results with the unorthodox tagged array sorting technique based on NVIDIA's Thrust library. GPU-ArraySort out-performs the tagged array sorting technique by sorting three times more data in a much smaller time. The developed tool and strategy is made available at https://github.com/pcdslab/GPU-ArraySort-2.0.

8. **MS-Reduce**

MS-Reduce is a linear-time tool that allows massive reduction in the amount of mass spectrometry data without significantly reducing the quality of the peptide deduction. Our novel data- reductive strategy for analysis of Big MS data is called MS-REDUCE and is capable of eliminating noisy peaks as well as peaks that do not contribute to peptide deduction before any peptide deduction is attempted. Our experiments have shown up to 100x speed up over existing state of the art noise elimination algorithms while maintaining comparable high quality matches. Using our approach we were able to process a million spectra in just under an hour on a moderate server which will be especially useful for processing in high-throughput environments. The algorithm has been implemented in Java and code/associated data sets are available on GitHub https://github.com/pcdslab/MSREDUCE.

9. **ParaDSRC**

ParaDSRC is a high-performance tool for compressing next generation sequencing data us- ing memory-distributed clusters. It uses domain decomposition and message passing inter- face (MPI) to distribute data on memory-distributed compute nodes. Our implementation gives near-linear speedups for most of the data sets with some evidence of super-linear speedups for some data sets. We report experimental results for up to 1 tera byte (TB). The algorithm has been implemented using C/C++ and MPI and the code is available on GitHub here https://github.com/PCDS/paraDSRC

10. PhosSA

PhosSA is a program for phosphorylation site assignment of LC-MS/MS data. It uses a linear-time and linear space dynamic programming strategy for phosphorylation site assignment. The algorithm optimizes the objective function defined as the summation of intensity peaks that are associated with theoretical peptide fragmentation ions. A classifier introduced in the algorithm exploits the specific characteristics of mass spectrometry data to distinguish between the correctly and incorrectly assigned site(s). The algorithm has been implemented in JAVA. An executable and instructions to use the software can be downloaded from http://helixweb.nih.gov/ESBL/PhosSA/.

11. CPhos

Cphos is a program to calculate and visualize evolutionary conserved Phosphorylation sites. CPhos utilizes an information theory-based algorithm to assess the conservation of phosphorylation sites among species. A conservation established from this approach can be used to potentially assess the functional significance of a particular phosphorylation site. A web- service and executable are available from http://helixweb.nih.gov/CPhos/

12. P-Pyro-Align

P-Pyro-Align is an open source parallel algorithm for multiple alignment of pyrosequencing reads from multiple genomes. The proposed alignment algorithm accurately aligns the erroneous reads and the accuracy of the alignment is confirmed from the consensus obtained from the multiple alignments. The algorithm uses domain decomposition for parallel computations of the local multiple alignments and a novel merging technique for global alignment of the reads. The proposed algorithm shows super-linear speedups for a large number of reads. Note that the algorithm is for multiple alignment of reads coming from different strains of genomes which cannot be handled using mapping of the reads to a reference genome. The code has been implemented using C/C++ and MPI libraries.

8. Student Research Supervision

8.1. Awards Received by Students and Postdocs

1. Muhammad Umair, *American Epilepsy Society Fellow*[10], Los Angeles CA, Sept 2024
2. Muhammad Umair, *Knight Foundation Fellowship in Healthcare Technology Innovation class*, Baptist Health Innovations, Baptist Health South Florida, August 2022
3. Muhammad Haseeb, *Best Graduate Research Award*, School of Computing & Information Sciences (SCIS), FIU, Dec 2021
4. Tianren Yang, *Best Graduate Teaching Award*, School of Computing & Information Sciences (SCIS), FIU, Dec 2020
5. Muaaz Awan, *All-University Graduate Research and Creative Scholar Award*, WMU (most significant honor that Western Michigan University bestows upon its graduate students), 2019.
6. Muaaz Awan, *WMU Graduate College Research Grant* for GPU computing, 2018;
7. Sandino N. Vargas-Pérez, *Winner of Regional 3MT competition*, 2017.
8. Muaaz Awan, *Department-Level Graduate Research and Creative Scholars Award* at WMU, 2015.

8.2. Postdoctoral Fellows Supervision

1. Zubair Saeed, PhD, April 2026 – to date
   Postdoctoral Fellow, KFSCIS FIU
2. Gabriel Bianchin de Oliveira, PhD, August 2025 – June 2026
   Postdoctoral Fellow, KFSCIS FIU
   First Position: Research Fellow at University of Campinas, Brazil
3. Muhammad Umair, PhD, Oct 2020 - August 2025
   Postdoctoral Fellow, KFSCIS FIU
   First position: Assistant Professor at Union College, NY
4. Mai A. Alduailij, PhD, Oct 2020 - Aug 2021
   Postdoctoral Research Fellow, KFSCIS FIU

[10] Selection as a Fellow of the American Epilepsy Society is an honor recognizing professional accomplishment and dedication in epilepsy. It is a way for AES to recognize members who have made meaningful contributions over time through research, clinical or academic achievement, volunteerism, and/or leadership.

First Position: Assistant Professor, College of Computer and Information Sciences (CCIS) Princess Nourah Bint Abdulrahman University, Riyadh, Saudi Arabia

7.1. PhD Dissertations (as Chair)

1. Fahad Almuqhim, Spring 2020 - Spring 2026
   Dissertation Title: Self-Supervised Representation Learning from Multi-Cohort Neuroimaging for Neurodevelopmental and Neurodegenerative Disorders
   First Position: Assistant Professor, Al Imam Mohammad ibn Saud Islamic University, Saudi Arabia
2. Usman Tariq, Fall 2018 - Fall 2023
   Dissertation Title: Novel Deep Learning Models for Mass Spectrometry-based Omics
   First Position: Research Scientist, Meta (formerly Facebook), Bay area CA USA
3. Oswaldo Artiles, Fall 2019 - Fall 2023
   Dissertation Title: Statistical and Machine Learning Analysis of the Human Brain Functional Network in a Multi-site Resting-state Functional MRI Database Framework
   First Position: Retired.
4. Muhammad Haseeb, Fall 2017 - Spring 2023
   Dissertation Title: High-Performance Computing Algorithms for Accelerating Peptide Identification from Mass-Spectrometry Data using Heterogeneous Supercomputers
   First Position: Postdoctoral Fellow, National Energy Research Scientific Computing Center (NERSC), Lawrence Berkeley (DOE) National Laboratory, Berkeley CA USA
   Last known position: Senior Software Engineer (IC4 level), NVIDIA.
5. Taban Eslami, Fall 2015 - Spring 2020
   Dissertation Title: High-Performance Computing and Machine-Learning Models for Big fMRI data
   First Position: Machine Learning and image processing engineer at ZEISS, Minneapolis Minnesota USA
   Last Known Position: ML Algorithm Validation Engineer at Apple, CA USA
6. Muaaz Gul Awan, Fall 2014 - Summer 2019
   Dissertation Title: High-Performance Computing Reductive Strategies for Big Data from LC-MS/MS based Proteomics
   First Position: Post-Doctoral Fellow and then permanent Staff Scientist, National Energy Research Scientific Computing Center, Lawrence Berkeley National Laboratory (Berkeley Lab), Berkeley CA USA
   Last Known Position: HPC Software Engineer, NVIDIA, CA
7. Sandino N. Vargas-Pérez, Summer 2014 - Fall 2017 (Defended Summer 2017) Dissertation Title: A Hybrid Parallel Approach to High-Performance Compression of Big Genomic Files and In-Compressor Data Processing
   First Position: Lecturer in CS department at WMU
   Last Known Position: Associate Professor, Kalamazoo College, Kalamazoo MI USA
8. Mohammed Aledhari, Fall 2014 - Fall 2017 (Defended Summer 2017)
   Dissertation Title: A Deep Learning-based Data Minimization Algorithm for Big Genomics Data in Support of IoT and Secure Smart Health Services
   Honors: Recipient of 2016 Graduate Student Research Grant at WMU for US$1000, Recipient of Gwen Frostic Doctoral Fellowship from Western Michigan University for US$ 4000
   First Position: Assistant Professor (Tenure-Track), Computer Science Department, Kenne- saw State University, Marietta, GA USA;
   Last Known Position: Assistant Professor (Tenure-Track) University of North Texas (UNT) in Fall 2022.
9. Mohammad Abu Shattal (jointly supervised with Prof. Ala Al-Fuqaha), Summer 2014 - Fall 2017 (Defended Fall 2017)
   Dissertation Title: Bio-Socially Inspired Strategies in Support of Dynamic Spectrum Access: An Evolutionary Game Theory Perspective
   First Position: Post-doctoral Fellow ECE department, Ohio State University, Columbus, OH Last Known Position: Senior Lecturer and Lead Faculty, Department of Computing Sciences and Mathematics, Franklin University, Columbus, OH USA

7.2. PhD Student Supervision

Current PhD students (post prelim)

1. N/A

Current PhD students (post qualifier)

1. Bilal Shabbir, Fall 2024 - Fall 2029 (expected)
   PhD student in the SCIS, FIU
2. Jazmin Lagier, Spring 2025 - Fall 2030 (expected)
   PhD student in the SCIS, FIU

Current PhD students (pre-qualifier)

3. Sumesh Kumar, Fall 2025 - Fall 2032 (expected)
   PhD student in the SCIS, FIU
4. Samuel Ebert, Spring 2023 - Spring 2028 (on-leave)
   PhD student in the SCIS, FIU
5. Robert Loredo, Summer 2025 - Fall 2030 (expected)
   PhD student in the SCIS, FIU
6. Avinash Jeewani, Spring 2026 – Fall 20230 (co-advised with Dr. Arif Sawrat)
   PhD student in the SCIS, FIU
7. Avash Palikhe, Spring 2026 – Fall 2031
   PhD student in the SCIS, FIU

Former PhD Students supervision

1. Sarah Rashidian, May 2012 to Aug 2012
   PhD Student at National Institutes of Health (NIH)
   Project Title: Phosphorylation site assignment for large-scale mass spectrometry data Last Known Position: PhD student at Towson University, MD USA

Member of PhD Dissertation Committee

1. Anwer Shees, FIU Electrical and Computer Engineering Department, 2026 - (Advisor: Arif Sarwat)
2. Uzair Khan, FIU Electrical and Computer Engineering Department, 2026 - (Advisor: Arif Sarwat)
3. Kasun Shashilaksha, FIU Electrical and Computer Engineering Department, 2024 - (Advisor: Arjuna Madanayake)
4. Md Adil Arman, FIU Biomedical Engineering Department, 2025 - (Advisor: Oleksii Shandra)
5. Md Abu Taher, FIU Electrical and Computer Engineering Department, 2024 - (Advisor: Arif Sarwat)
6. Hasan Iqbal, FIU Electrical and Computer Engineering Department, 2024 - (Advisor: Arif Sarwat)
7. Seyedsina Nabavirazavi, FIU School of Computing, 2024 - (Advisor: SS Iyengar)
8. Azwad Anjum Islam, FIU School of Computing, 2024 - (Advisor: Mark A. Finlayson)
9. Emmanuel Garcia, FIU School of Computing, 2024 - (Advisor: Mark A. Finlayson)
10. Sukanta Roy, FIU Electrical and Computer Engineering Department, 2024 - (Advisor: Arif Sarwat)
11. Pritom Kumar Saha, FIU Biomedical Engineering Department, 2024 - (Advisor: Oleksii Shandra)
12. Dabasish Saha, FIU Biomedical Engineering Department, 2024 - (Advisor: Nikolaos Tsoukias)
13. Milad Behnamfar, FIU Electrical and Computer Engineering Department, 2023 - (Advisor: Arif Sarwat)
14. Andrew Forero, FIU Department of Chemistry, 2023 - (Advisor: Francisco Lima)
15. Anshu Sharma, FIU School of Computing, 2023 - (Advisor: Mark Finlayson)
16. Joseph Stevenson Alexander, FIU Department of ECE, 2023 - (Advisor: Arif Sarwat)
17. Khandaker Mamun Ahmed, FIU School of Computing, 2023 - (Advisor: Hadi Amini)
18. Seyedsina Nabavirazavi, FIU School of Computing, 2023 - (Advisor: SS Iyengar)
19. Tisa Islam Erana, FIU School of Computing, 2022 - (Advisor: Mark Finlayson)
20. Masrur Sobhan, FIU School of Computing, 2022 - (Advisor: Ananda M. Mondal)
21. Samira Zad, FIU School of Computing, 2019 -2022 (Advisor: Mark A. Finlayson)
22. Abdullah Al Mamun, FIU School of Computing, 2019 - 2022 (Advisor: Ananda M. Mondal)
23. Raihanul Bari Tanvir, FIU School of Computing, 2019 - 2023 (Advisor: Ananda M. Mondal)

24. Adnan Maruf, FIU School of Computing, 2020 - 2023 (Advisor: Janki Bhimani)
25. Yashas Hariprasad, FIU School of Computing, 2022 - (Advisor: SS Iyengar)
26. Samuel A. Miller, FIU Department of Chemistry, 2021 - (Advisor: Francisco AF Lima)
27. Hugo Riggs, FIU Electrical and Computer Engineering, 2021 - 2023 (Advisor: Arif Sarwat)
28. Anurag Acharya, FIU School of Computing, 2019 - (Advisor: Mark A. Finlayson)
29. Mohammad Asif Iqbal Khan, FIU Electrical and Computer Engineering, 2020 - 2022 (Advisor: Sumit Paudyal)
30. Morgan Jusko, FIU Department of Psychology, 2020 - (Advisor: Joseph Raiker/Gregory Fabiano)
31. Pedro Espina, FIU School of Computing, 2020 - (Advisor: Niki Pissinou)
32. Manoj Pravakar Saha, FIU School of Computing, 2020 - (Advisor: Janki Bhimani)
33. Daniel Ruiz-Perez, FIU School of Computing, 2019 - (Advisor: Giri Narasimhan)
34. Liqun Yang, FIU School of Computing, 2019 - (Advisor: Wei Zeng)
35. Mohammed Aldawsari, FIU School of Computing, 2019 - (Advisor: Mark A. Finlayson)
36. Taylor Salo, FIU Department of Psychology, 2018 - (Advisor: Angela Laird)
37. Semsi Coskun, WMU Civil Engineering Department, 2017 - (Advisor: Houssam A Toutanji)
38. Hasnaa Al Shaikhli, WMU CS department, 2017 - (Advisor: Elise DeDoncker)
39. Samah Rahamneh, WMU ECE department, 2017 - (Advisor: Ikhlas Abdel-Qader)
40. Ting-Yu Mu, WMU CS department, 2017 - (Advisor: Ala Al-Fuqaha)
41. Omar Abed Darwish, WMU CS department, 2017 - (Advisor: Ala Al-Fuqaha)
42. Chung-Ling Lin, WMU CS Department, 2014 - (Advisor: Wuwei Shen)

Member of PhD Qualifying-Exam Committee

1. Jimeng Shi, FIU School of Computing, 2023 - (Advisor: Giri Narasimhan)
2. Yashas Hariprasad, FIU School of Computing, 2023 - (Advisor: S.S. Iyengar)
3. Anshu Sharma, FIU School of Computing, 2023 - (Advisor: Mark Finlayson)
4. Masrur Sobhan, FIU School of Computing, 2022 - (Advisor: Ananda Mondal)
5. Tisa Islam Erana, FIU School of Computing, 2022 - (Advisor: Mark Finlayson)
6. Robin Stubbs, Department of Biological Sciences, 2021 (Advisor: DeEtta (Dee) Mills)
7. Tasmia Aqila, FIU School of Computing, 2020 - (Advisor: Ananda M. Mondal)
8. Abdullah Al Mamun, FIU School of Computing, 2019 - (Advisor: Ananda M. Mondal)
9. Anurag Acharya, FIU School of Computing, 2019 - (Advisor: Mark A. Finlayson)
10. Arpit Mehta, FIU School of Computing, 2018 - (Advisor: Giri Narasimhan)
11. Labiba Jahan, FIU School of Computing, 2018 - (Advisor: Mark A. Finlayson)

7.3. M.S. Student Supervision

Current MS Thesis Students

1. N/A

Current Non-Thesis Research Students

1. Bronwyn Brett (CIS, MS), Summer 2025 - to date

Former MS Thesis Students

1. Usman Tariq (ECE, WMU), Fall 2016 - Spring 2018
   Thesis Title: Power-Efficient and Highly Scalable Parallel Graph Sampling using FPGAs Last Known Position: PhD student WMU
2. Ansab Ali (ECE, IIT Chicago co-advised with Ashfaq Khokhar), Fall 2014 - Spring 2016 Thesis Title: Clustering Algorithm for Mass Spectrometry data using general-purpose com- puting on graphics processing units (GPU's)
   Last Known Position: Engineer at Intel Corp., Portland OR
3. Dana Emad Abdul Qader (ECE, WMU), Fall 2014 - Fall 2015
   Thesis Title: A High Performance Architecture for an Exact Match Short-Read Aligner Using Burrow-Wheeler Transform on FPGAs
   Last Known Position: QA Engineer at Stryker Corporation, San Jose CA

Former Non-Thesis MS Research Students

1. Jeisson Parra (CIS, MS), Sept 2025 to date

Capstone project: Multimodal Deep Learning for Early Detection and Classification of Brain Tumors Using MRI and CT scans

1. Brianna Valdes (CIS, MS), Sept 2025 - to date

Capstone Project: Deep Learning for identifying Circulating extracellular vesicles (EVs) specific to multiple sclerosis

2. Ulysses Echeverria (CIS, MS), Nov 2024 – Summer 2025 Machine-Learning models for PET scans related to ADRD
First position: PhD student in KFSCIS FIU.
3. Myrah Bisht (CIS, MS), August 2024 – to date
Project: Creating mass spectrometry-based omics benchmarks
First position: not known
4. Paras Parani (CIS, MS), May 2024 – Summer 2025
Research project on models for epilepsy seizure prediction
First Position: Software Engineer at Amazon, Seattle WA
5. Syed Talha Khalid (CIS, MS), May 2024 – Summer 2025
Project: Cloud computing infrastructure for MS data omics
First Position: Full Stack Engineer at AI-NeoTech LLC Miami Florida
6. Tianren Yang (CS, FIU), Fall 2020 - Fall 2023
Last known position: N/A
7. Sumesh Kumar (CS, FIU), Fall 2019 - Fall 2023
Last Known Position: Hardware engineer at HP
8. Abhishek Bhattarai, (CS, MS), Aug 2023 – Dec 2023
MS research project for epilepsy, FIU
Last known position: Full stack engineer at AI-NeoTech LLC
9. Samuel Ebert, (CS, MS), Dec 2022 - May 2023
Last known position: Admitted to PhD program KFSCIS FIU, summers 2023
10. Siddhi Lanke (CIS, MS), Aug 2022 - June 2023
MS data science student for Capstone Project, FIU
11. Abhilash Rana (CIS, MS), Aug 2022 - June 2023
MS data science student for Capstone Project, FIU
12. Carlos Escudero Castillo (CIS, MS), Aug 2022 - June 2023
MS data science student for Capstone Project, FIU
13. Harun Oz (CIS, FIU), August 2019 - Dec 2019 MS student in SCIS FIU
14. Priyanka Priyanka, August 2018 - Oct 2018
MS data science student for Capstone Project, FIU
15. Blake Wrege, Spring 2017 - Summer 2017 MS student in CS department at WMU
Project Title: Graphical user interfaces for networking solutions for big data Last position: Site Reliability Engineer (SRE) at Apple
16. Srikanth Aravamuthan, Fall 2015 - Summers 2016 MS student in Statistics department at WMU
Project Title: statistical models for mass spectrometry-based proteomics data Last position: N/A
17. Alan Perez-Rathke, Jan 2009 to Jan 2010 MS student at University of Illinois at Chicago
Project Title: Multiple sequence alignment algorithms using HPC systems for pyrosequencing reads
Last Known Position: MD student at University of Illinois at Chicago
18. Jarek Gwarnicki, Jan 2009 to Jan 2010
MS student at University of Illinois at Chicago
Project Title: Quality assessment of MSA of pyrosequencing reads using domain decompo- sitions.
Last Known Position: Software Engineer at a Chicago based gaming company

7.4. Undergraduate Researchers

Current Undergraduate Research Students

1. Anju Singh, Jan 2025 - to date
   B.Sc student in the CIS Department, FIU

Former Undergraduate Research Students

1. Jalen Atwell, Feb 2024 - May 2024
   B.Sc student in the CIS Department, FIU
2. Hermes Bonilla, June 2021 - August 2023

B.Sc student in the CIS Department, FIU
3. Lazaro Hurtado, Jan 2022 - August 2022
   B.Sc student in the CIS Department, FIU
4. Paulina Acosta, Jan 2022 - August 2022
   B.Sc student in the CIS Department, FIU
   Last known position: Software engineer at Apple.
5. Osazomon Imarenezor, Jan 2021 - May 2021
   B.Sc student in the CIS Department, FIU
6. John Quitto Graham, Jan 2019 - June 2020
   B.Sc student in the CIS Department, FIU
7. Alejandra Vasquez, June 2020 - June 2021
   B.Sc student in the CIS Department, FIU
   Last known position: Software Engineer at multinational company
8. Emily Costa, Sept 2019 - Aug 2020
   B.Sc student in the CIS Department, FIU
   Last known position: PhD student at Northeastern University
9. Fatima Afzali, Sept 2018 - August 2019
   B.Sc student in the CIS and Mathematics Department, FIU
   Last known position:
10. Richard Larancuente, August 2018 - Dec 2018
    B.Sc student in the CIS Department, FIU
11. Alexis Gonzalez, August 2018 - Dec 2018
    B.Sc student in the CIS Department, FIU
12. Alexander S. Cadigan, May 2017 - Aug 2017
    B.Sc. student in CS Department, Kalamazoo College
    Project Title: Scalability study of Open-Search Algorithms for MS based Proteomics data Last Known Position: Computer Science undergrad student at Kalamazoo College, Kalama- zoo MI
13. James A Novorita, August 2017 - Dec 2017
    B.Sc student in the ECE Department, WMU
    Project Title: k-mer based alignment of big genomic data using FPGA's Last Known Position: Engineer at Stryker Corporation, Kalamazoo MI
14. Blake Wrege, Summer 2016 - Fall 2016
    B.Sc student in the CS Department, WMU
    Project Title: Graphical User interfaces for big data transmission and sharing
    Last Known Position: MS student at WMU
15. Melissa Basileo, Summer 2015 - Summer 2015
    B.Sc student in the CS Department, WMU
16. Akshay Sanghi, May 2013 to Aug 2013
    B.Sc. student at National Institutes of Health (NIH)
    Project Title: A Knowledge Base of Vasopressin Actions in Kidney Last Known Position: MD student at Johns Hopkins University
17. Boyang Zhao, Dec 2011 to Aug 2012
    B.Sc. student at National Institutes of Health (NIH)
    Project Title: Assessing the Functional Relevance of Phosphorylation Sites using Information Theory and Naive Bayes Classifications
    Last Known Position: PhD student at Massachusetts Institute of Technology (MIT)
18. Jacqueline Douglass, Sept 2010 to Aug 2011
    B.Sc. student at National Institutes of Health (NIH)
    Project Title: Protein Kinase Target-Sequence Profiling Using LC-MSMS Last Known Position: MD/PhD student at Johns Hopkins University

7.5. High School Research Students

Current High School Research Students

1. N/A

Former High School Research Students

- Hermes Bonilla, June 2020 - June 2021 John

Ferguson High School
Last Known position: Undergrad student at SCIS FIU

- Zoe Srackangast, Gwen Park, and Isabel Holton, Sept 2017 - March 2018 High School student at Kalamazoo Area Mathematics & Science Center
Project Title: Development of Algorithm for Classification of Proteins found in Environmental Samples (Recipient of Out Standing Achievement for Ability and Creativity Award @ SW-Michigan International Science & Engineering Fair (ISEF) 2018)
Last Known Position: Isabel Holton joined Undergraduate program in Electrical Engineering at University of Michigan Ann Arbor
- Lily Kitagawa, Oct 2015 - March 2016
High School student at Kalamazoo Area Mathematics & Science Center
Project Title: Counting kmers in big genomic data (Recipient of Intel Excellence in Computer Science Award 2016)
Last Known Position: Undergraduate student (Computer Science) at California Institute of Technology (Caltech)
- Binh Le, August 2014 - March 2015
High School student at Kalamazoo Area Mathematics & Science Center
Project Title: GUI development of computational biology tools for domain scientists
Last Known Position: Undergraduate student at Massachusetts Institute of Technology (MIT)
- Adam Loles, August 2014 - March 2015
High School student at Kalamazoo Area Mathematics & Science Center
Project Title: GUI development of computational biology tools for domain scientists Last Known Position: Undergraduate student at Colgate University.

7.6. External Examiner

- Umair Ayub, Department of Computer Science, National University of Computer and Emerging Sciences, Islamabad, Pakistan Sept 2022 (Advisor: Dr. Hammad Naveed)
- Kanzal Iman, Department of Biology, SBA School of Science and Engineering, Lahore University of Management Sciences (LUMS), Lahore, Pakistan, 2022 (Advisor: Dr. Safee Ullah Chaudhary)

8. Professional and Academic Service

8.1. Editorial Boards

- **Associate Editor***, IEEE Transactions on Computational Biology and Bioinformatics (TCBB), 2025 to date*
- **Editor**, *PLOS Mental Health*, Oct 2023 to date
- **Associate Editor**, *Experimental Results*, Cambridge University Press, April 2022
- **Review Editor** on the Editorial Board of *Digital Public Health* (specialty section of Frontiers in Public Health, Frontiers in ICT and Frontiers in Computer Science), July 2021 to date
- **Statistics and Bioinformatics Board** for Journal of the American Society of Nephrology (JASN), Feb 2019 to date
- **Associate Editor** for Springer Journal of Network Modeling Analysis in Health Informatics and Bioinformatics, Jan 2014 to date.

8.2. Guest Editor Special Issues

- Guest Editor for *Frontiers in Psychiatry*, Jan 2024 (one paper assignment)
- Guest Editor for *Journal of Bioinformatics and Computational Biology (JBCB)* special issue containing selected papers from the 7th International Conference on Bioinformatics and Computational Biology, Feb 2016 (with Nurit Haspel and Hisham Al-Mubaid).
- Guest Editor for *Journal of Bioinformatics and Computational Biology (JBCB)* special issue containing selected papers from the 6th International Conference on Bioinformatics and Computational Biology, Oct 2014 (with Hisham Al-Mubaid and Bhaskar Dasgupta).
- Guest Editor for *Journal of Bioinformatics and Computational Biology (JBCB)* special issue containing selected papers from the 5th International Conference on Bioinformatics and Computational Biology, Oct 2013 (with Hisham Al-Mubaid and Bhaskar Dasgupta).

8.3. Conference/Workshop Chair

- **Workshop Co-Chair**, 6th Workshop on Scalable Machine Learning for Health and Biomedical Data (SML-HBD) (previously HPC-BOD), in conjunction with IEEE BIBM, Dec 2025
- **Workshop Co-Chair**, 5th Workshop on High Performance Computing for Big Omics Data (HPC-BOD), in conjunction with IEEE Big Data, Dec 2024
- **Workshop Co-Chair**, 22nd IEEE International Workshop on High Performance Computational Biology (HiCOMB), in conjunction with IEEE International Parallel and Distributed Pro- cessing Symposium (IPDPS), St. Petersburg Florida, USA, May 15th, 2023
- **Session Chair** for Big Data and AI, IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'22) joinly organized with the 17TH IEEE-EMBS International Conference on Wearable and Implantable body Sensor Networks (BSN'22), Sept 2022
- **Workshop Co-Chair**, 4th Workshop on High Performance Computing for Big Omics Data (HPC-BOD), in conjunction with IEEE BIBM, Dec 2022
- **Workshop Co-Chair**, 3rd Workshop on High Performance Computing for Big Omics Data (HPC-BOD), in conjunction with ACM-BCB, Aug 2021
- **Workshop Co-Chair**, Workshop on Machine-Learning, and High-Performance Computing for Proteomics, in conjunction with KFSCIS and LUMS, Sept 2021
- **Workshop Co-Chair**, 2nd Workshop on High Performance Computing for Big Omics Data (HPC-BOD), in conjunction with ACM-BCB, Oct 2020
- **Workshop Chair**, 1st IEEE Workshop on High Performance Computing for Big Omics Data (IEEE HPC-BOD), in conjunction with IEEE BIBM, Nov 2019
- Track Chair for Mathematical Modeling and Scientific Computing, IEEE International Confer- ence on Advances in Computing, Communications and Informatics (ICACCI), Jaipur India, 2016
- Session Chair, IEEE Workshop on High Performance Computing for Big Data Computational Biology (IEEE HPC-BCB), in conjunction with IEEE BIBM, Washington DC, Nov 2015
- **Founding Workshop Chair**, 1st IEEE Workshop on High Performance Computing for Big Data Computational Biology (IEEE HPC-BCB), in conjunction with IEEE BIBM, Nov 2015
- Tutorials Co-Chair for IEEE International Conference on Collaboration Technologies and Systems (CTS) Atlanta, Georgia, USA, May 2015
- **Program Co-Chair** for 7th Bioinformatics and Computational Biology (BICoB) Conference, March 2015
- Tutorials Co-Chair for IEEE International Conference on Collaboration Technologies and Systems (CTS) Minneapolis, Minnesota, USA, May 2014
- **Program Co-Chair** for International Workshop on Algorithms for Computational Biology (ACB), 2014
- **Program Co-Chair** for 6th Bioinformatics and Computational Biology (BICoB) Conference, Las Vegas, Nevada, March 24-26, 2014
- Session Chair, International workshop on computational proteomics, IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Oct 2012
- Session Chair, IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Oct 2012
- **Program Co-Chair** for 5th Bioinformatics and Computational Biology (BICoB) Conference, Honolulu, Hawaii, March 4-6, 2013
- **Program Co-Chair** for Bioinformatics and Computational Biology (BICoB) Conference, Las Vegas, Nevada, USA, March 12 - 14, 2012

8.4. Selected Conference/Workshop Program Committee Membership

1. Program committee member, 35th ACM International Conference on Information and Knowledge Management (CIKM 2026), Rome Italy, 2026
2. Program committee member, 12th IEEE International Conference on Collaboration and Internet Computing (IEEE CIC 2026), San Jose, California, November 2026
3. Program committee member, 29th International Conference on Medical Image Computing & Computer Assisted Interventions (MICCAI), Abu Dhabi, 4-8 Oct 2026
4. Inaugural Program committee member, ACM Conference on Knowledge Discovery and Data Mining, AI4Sciences Track, (ACM KDD) 2026, Jeju, Korea, August 9-13, 2026
5. Program committee member, IEEE International Conference on BigData (IEEE BigData), Phoenix, AZ, USA, Dec 14-17, 2026

6. Program committee member, IEEE 23rd International Symposium on Biomedical Imaging (ISBI), London UK, April 8-11, 2026
7. Program committee member, SoutheastCon, Huntsville, Alabama USA, March 13-15 2026
8. Program committee member, Eleventh IEEE International Conference on Collaboration and Internet Computing (IEEE CIC), Pittsburgh, USA, Nov.11-14, 2025.
9. 33rd ACM Conference on Information and Knowledge Management (CIKM), Seoul, South Korea, Nov 10-14, 2025
10. Program committee member, 2025 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM), Wuhan, China, December 15-18, 2025.
11. Program committee member for IEEE International Workshop on High Performance Computational Biology (HiCOMB), International Parallel & Distributed Processing Symposium (IPDPS), Milan Italy, June 4th 2025
12. Program committee member, The 10th IEEE International Conference on Collaboration and Internet Computing (IEEE CIC), October 28 - 30, 2024, Washington D.C. USA
13. Program committee member, The 33rd ACM International Conference on Information and Knowledge Management (CIKM), 21-25 Oct, 2024, Boise ID USA
14. Program committee member, 2024 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM), Portugal, Dec 2024
15. Program committee member, IEEE BigData, Washington DC, Dec 15 to Dec 18, 2024
16. Program committee member, IEEE SoutheastCon, Atlanta GA, USA on March 15-17, 2024
17. Program committee member, 44th IEEE International Conference on Distributed Computing Systems (ICDCS), Jersey City, New Jersey, USA on July 16-19, 2024
18. Program committee member, 2nd Quantum Computing Algorithms, Systems, and Applications (Q-CASA), IPDPS San Francisco, California on May 27-31, 2024
19. Program committee member, The 32nd ACM International Conference on Information and Knowledge Management (CIKM), 21-23 Oct, 2023, University of Birmingham and Eastside Rooms, UK
20. Program committee member, IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM) December 5-8, 2023, Istanbul, Turkey
21. Program committee member, 1st Quantum Computing Algorithms, Systems, and Applications (Q-CASA), IPDPS St. Petersburg, Florida on May 15, 2023.
22. Program committee member, 14th Student Research Symposium (SRS), 29th IEEE International Conference on High Performance Computing, Data, & Analytics (HIPC 2022) Decem- ber 18-21, 2022 Bengaluru, INDIA
23. Senior Program Committee member, ACM International Conference for Information and Management (CIKM2022), Atlanta GA, Oct 17 2022
24. Program committee, 28th IEEE INTERNATIONAL CONFERENCE ON HIGH PERFORMANCE COMPUTING, DATA, & ANALYTICS (HiPC), Dec 2021
25. Program committee member for Advanced Machine Learning and Applications: Federated Learning and Meta-Learning (AML-IoT FLAME 2020) at ICMLA 2020
26. Program committee member for 15th International Wireless Communications & Mobile Computing Conference (IWCMC 2019), Tangier, Morocco, June 24-28, 2019
27. Program committee member for IEEE International Workshop on High Performance Computational Biology (HiCOMB), International Parallel & Distributed Processing Symposium (IPDPS), Rio de Janeiro, Brazil, May 20, 2019
28. Program committee member, Fourth International Conference on Research in Computational Intelligence and Communication Network (ICRCICN 2018), Kolkata, India, November 22 - 23, 2018
29. Program committee member, Symposium on Bioinformatics and Bioforensics (SBB'18), ICACCI, Bangalore India, Sept 2018
30. Program committee member, Workshop on Large Scale Computational Physics (LSCP), International Conference on Computational Science and Applications (ICCSA), June 2018
31. Program committee member, First International Symposium on Signal and Image Processing (ISSIP-2017), Kolkata, India Nov 2017
32. Program committee member, IEEE International Conference on Advanced Computational and Communication Paradigms (ICACCP-2017), Sikkim, India 8-10th September 2017

33. Program committee member, 10th International Conference on Security, Privacy and Anonymity in Computation, Communication and Storage (SpaCCS 2017), Guangzhou, China, December 12-15, 2017.
34. Program committee member, International Symposium on Bioinformatics and Bio forensics (SBB'17), International Conference on Advances in Computing, Communications and Informatics (ICACCI), Manipal, India September 13-16, 2017
35. Program Committee Member, International Conference on Advances in Computing, Communications and Informatics (ICACCI), Jaipur, India 2017
36. Program committee member, International Symposium on Bioinformatics and Bio forensics (SBB'16), International Conference on Advances in Computing, Communications and Informatics (ICACCI), Jaipur, India September 21-24, 2016
37. Program committee member, International Symposium on Network Enabled Health Informatics, Biomedicine and Bioinformatics (HI-BI-BI), Aug 2016
38. Program committee member, Workshop on Large Scale Computational Physics (LSCP), International Conference on Computational Science and Applications (ICCSA), June 2016
39. Program committee member, The 15th IEEE International Conference on Data Mining (ICDM), 2015
40. Program committee member, IEEE Symposium on Signal and Information Processing for Software-Defined Ecosystems and Green Computing (IEEE GlobalSIP), Orlando Florida Dec 2015
41. Program committee member, Advancement in Petroleum and Chemical Engineering Technology and Applications International Conference (APCETA), Dec 2015
42. Program committee member, IEEE Information Reuse and Integration in Health Informatics (IRI-HI), San Francisco CA August 2015
43. Program committee member, IEEE/ACM International Symposium on Network Enabled Health Informatics, Biomedicine and Bioinformatics (HI-BI-BI), Paris, France, Aug 26-27, 2015
44. Program committee member, International Workshop on Algorithms for Computational Biology (ACB-2014), Delhi, India, September 2014
45. Program committee member, IEEE/ACM International Symposium on Network Enabled Health Informatics, Biomedicine and Bioinformatics (HI-BI-BI), Beijing, China, August 18-19, 2014
46. Member Steering Committee, 2nd Computational Science Conference, Pakistan Society of Computational Science and Biology (PSCS/PSCB) for High School Students, Islamabad Pakistan, Oct 2013
47. Program committee member for IEEE International Workshop on High Performance Computational Biology (HiCOMB), International Parallel & Distributed Processing Symposium (IPDPS), May 27, 2013
48. Program committee member for ACM conference on Bioinformatics, Computational Biology and Biomedicine (ACM-BCB) conference, Orlando Florida, Oct 7-10, 2012
49. Program committee member for 11th International Workshop on Data Mining in Bioinformatics (BIOKDD '12)
50. Program committee member for 8th International Conference on Bioinformatics and Genome Regulation and System Biology (BGRS/SB) 2012

8.5. Peer Reviewer (Selected partial list)

1. Peer Reviewer for IEEE Transactions on Systems, Man and Cybernetics: Systems, June 2026
2. Peer Reviewer for Journal of Biomedical and Health Informatics - JBHI, May 2026
3. Peer Reviewer for IEEE Transactions on Neural Networks and Learning Systems, Dec 2025
4. Peer Reviewer for ACS Journal of Proteome Research, Nov 2025
5. Peer Reviewer for Critical Reviews in Biomedical Engineering, Nov 2025
6. Peer Reviewer for Result in Chemistry, Nov 2025
7. Peer Reviewer for IEEE Transactions on Neural Networks and Learning Systems, Nov 2025
8. Peer Reviewer for IEEE Transactions on Modeling and Computer Simulation, Sept 2025
9. Peer Reviewer for IEEE Transactions on Neural Networks and Learning Systems, August 2025
10. Peer Reviewer for Journal of Neuroscience Methods, August 2025
11. Peer Reviewer for ACM Transactions on Computing for Healthcare, July 2025
12. Peer Reviewer for *Epilepsy & Behavior*, May 2025

13. Peer Reviewer for *Oxford Bioinformatics*, Jan 2025
14. Peer Reviewer for *Oxford Bioinformatics*, Nov 2024
15. Peer Reviewer for *Nature Computational Science*, Oct 2024
16. Peer Reviewer for *Computer Science Review*, August 2024
17. Peer Reviewer for *Computers in Biology and Medicine*, July 2024
18. Peer Reviewer for *Oxford Bioinformatics*, July 2024
19. Peer Reviewer for *Journal of Proteome Research*, April 2024
20. Peer Reviewer for *Heliyon Cell Press*, April 2024
21. Peer Reviewer for *Nature Translational Psychiatry*, Jan 2024
22. Peer Reviewer for *ACS Journal of Proteome Research*, Jan 2024
23. Peer Reviewer for *Computers in Biology and Medicine*, Jan 2024
24. Peer Reviewer for *Computers in Biology and Medicine*, Nov 2023
25. Peer Reviewer for *Journal of Parallel and Distributed Computing (JPDC)*, Nov 2023
26. Peer Reviewer for *Wiley Psychiatry and Clinical Neurosciences*, Nov 2023
27. Peer Reviewer for *NeuroImage*, Nov 2023
28. Peer Reviewer for *Psychiatry Research: Neuroimaging*, Oct 2023
29. Peer Reviewer for *Neuroscience Applied*, Oct 2023
30. Peer Reviewer for *Nature Translational Psychiatry*, August 2023
31. Peer Reviewer for *BMC Bioinformatics*, August 2023
32. Peer Reviewer for *Heliyon Cell Press*, August 2023
33. Peer Reviewer for *Oxford Bioinformatics*, August 2023
34. Peer Reviewer for *Heliyon Cell Press*, August 2023
35. Peer Reviewer for *Transactions on Neural Systems & Rehabilitation Engineering*, July 2023
36. Peer Reviewer for *Computer Methods and Programs in Biomedicine*, July 2023
37. Peer Reviewer for *Hyliyon Cell Press*, July 2023
38. Peer Reviewer for *Elsevier Neuroscience Informatics*, July 2023
39. Peer Reviewer for *Elsevier Asian Journal of Psychiatry*, June 2023
40. Peer Reviewer for *IEEE Transaction on on Cybernetics*, March 2023
41. Peer Reviewer for *Behavioural Brain Research*, Jan 2023
42. Peer Reviewer for *Cybernetics and Systems*, Dec 2022
43. Peer Reviewer for *Nature Scientific Reports*, Nov 2022
44. Peer Reviewer for *Computer Methods and Programs in Biomedicine*, Sept 2022
45. Peer-reviewer for *IAES International Journal of Artificial Intelligence (IJ-AI)*, Sept 2022.
46. Peer Reviewer for *Wiley Psychophysiology*, Sept 2022
47. Peer Reviewer for *Computer Methods and Programs in Biomedicine*, July 2022
48. Peer Reviewer for *Journal of King Saud University - Computer Sciences*, May 2022
49. Peer Reviewer for *IEEE Journal of Biomedical and Health Informatics*, March 2022
50. Peer Reviewer for Journal of the *American Society of NEPHROLOGY*, March 2022
51. Peer Reviewer for Oxford *Bioinformatics*, Feb 2022
52. Peer Reviewer for *IEEE Transactions on Medical Imaging*, August 2021
53. Peer Reviewer for *ACM Computing Surveys*, Aug 2021
54. Peer Reviewer, *Wiley Computational and Systems Oncology*, Dec 2020
55. Peer Reviewer for *ACM Computing Surveys*, Dec 2020
56. Peer Reviewer for *Frontiers Neurology*, Nov 2020
57. Peer Reviewer for *IEEE Transactions on Cloud Computing*, Oct 2020
58. Peer Reviewer for Oxford *Bioinformatics*, Oct 2020
59. Peer Reviewer for Journal of the *American Society of NEPHROLOGY*, Sept 2020
60. Peer Reviewer for *Plos One*, August 2020
61. Peer Reviewer for Oxford *Bioinformatics*, June 2020
62. Peer Reviewer for *IEEE Access*, June 2020
63. Peer Reviewer for Journal of the *American Society of NEPHROLOGY*, May 2020
64. Peer Reviewer for *Springer Journal Network Modeling Analysis in Health Informatics and Bioinformatics (NHIB)*, May 2020
65. Peer Reviewer for *ACM Transactions on Internet of Things*, April 2020
66. Peer Reviewer for *BMC Bioinformatics*, March 2020
67. Peer Reviewer for Springer Nature *Translational Psychiatry*, Feb 2020

68. Peer Reviewer for MDPI *Journal of Clinical Medicine*, Jan 2020
69. Peer Reviewer for Elsevier *Biocybernetics and Biomedical Engineering*, Nov 2019
70. Peer Reviewer for *IEEE Access*, Nov 2019
71. Peer Reviewer for Oxford *Bioinformatics*, Oct 2019
72. Peer Reviewer for Journal of the *American Society of NEPHROLOGY*, Oct 2019
73. Peer Reviewer for Oxford *Bioinformatics*, Sept 2019
74. Peer Reviewer for Oxford *Bioinformatics*, August 2019
75. Peer Reviewer for Journal of the *Evolutionary Biology*, July 2019
76. Peer Reviewer for Journal of the *American Society of NEPHROLOGY*, May 2019
77. Peer Reviewer for *BMC Bioinformatics*, April 2019
78. Peer Reviewer for Oxford *Bioinformatics*, April 2019
79. Peer Reviewer for *Journal of Parallel and Distributed Computing (JPDC)*, Jan 2019
80. Peer Reviewer for *BMC Bioinformatics*, Jan 2019
81. Peer Reviewer for Journal of the *American Society of NEPHROLOGY*, Jan 2019
82. Peer Reviewer for *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, Dec 2018
83. Expert reviewer for book proposal for *Springer Biomedicine*, August 2018
84. Peer Reviewer for *BMC Bioinformatics*, August 2018
85. Peer Reviewer for *BMC Bioinformatics*, July 2018
86. Peer Reviewer for *Springer Signal, Image and Video Processing Journal*, July 2018
87. Peer Reviewer for *IEEE Transactions on Parallel and Distributed Systems*, Feb 2018
88. Peer Reviewer for *Wiley Proteomics*, Nov 2017
89. Peer Reviewer for *Wiley Proteomics*, Oct 2017
90. Peer Reviewer for *BMC Bioinformatics*, Sept 2017
91. Peer Reviewer for *Wiley Proteomics*, Aug 2017
92. Peer Reviewer for *IEEE Transactions on Parallel and Distributed Systems*, Aug 2017
93. Peer Reviewer for *Software Quality Journal*, Jan 2017
94. Peer Reviewer for *Springer Journal Network Modeling Analysis in Health Informatics and Bioinformatics (NHIB)*, Jan 2017
95. Peer Reviewer for Oxford *Bioinformatics*, Nov 2016
96. Peer Reviewer for *Nature Protocols*, Oct 2016
97. Peer Reviewer for Oxford *Bioinformatics*, Sept 2016
98. Peer Reviewer for *Springer Journal Network Modeling Analysis in Health Informatics and Bioinformatics (NHIB)*, Sept 2015
99. Peer Reviewer for *Journal of Bioinformatics and Computational Biology (JBCB)*, July 2015
100. Peer Reviewer for *Journal of Computer Science and Technology (JCST)*, May 2015
101. Peer Reviewer for *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, March 2015
102. Peer Reviewer for *Springer Journal Network Modeling Analysis in Health Informatics and Bioinformatics (NHIB)*, Feb 2015
103. Peer Reviewer for *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, Feb 2015
104. Reviewer for Oxford *Bioinformatics*, Jan 2015
105. Reviewer for *International Journal of Information and Communication Technology (IJICT)*, Jan 2015
106. Reviewer for *International Journal of Computers and Their Applications (IJCA)*, September 2014
107. Peer Reviewer for *Springer Journal Network Modeling Analysis in Health Informatics and Bioinformatics (NHIB)*, March 2014
108. Peer Reviewer for *International Journal of Computer Systems Science and Engineering (IJC-SSE)*, 2013
109. Peer Reviewer for *BMC Proteome Science*, March 2013
110. Peer Reviewer for *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2012
111. Peer Reviewer for *International Journal of Computer Systems Science and Engineering (IJC-SSE)* 2009 to 2011.

8.6. Proposal Panels and Expert Reviews

1. Expert Reviewer, Swiss National Science Foundation, June 2026
2. NSF Panelist, OAC Research Experiences for Undergraduates (REU) Site, Jan 2026
3. NIH Study Section Reviewer, Director's New Innovator Award Program (DP2), Center for Scientific Review (CSR), National Institutes of Health, Dec 2025
4. NIH Study Section Panelist, Biodata Management and Analysis (BDMA) study section, Center for Scientific Review (CSR), National Institutes of Health, Jan 2026
5. NIH study section reviewer, NIH Maximizing Investigators' Research Award (MIRA MCST-G (56)) (R35) panel, July 2025
6. External Reviewer for Faculty Tenure, Department of Computer Sciences and Engineering (FCSE), Ghulam Ishaq Khan (GIK) Institute of Engineering Sciences & Technology, Topi Pakistan, Sept Dec 2024
7. NIH study section reviewer, NIH Director's New Innovator Award Program (DP2), Dec 2024
8. NIH Study Section Panelist, Clinical Data Management and Analysis Study Section (CDMA), Center for Scientific Review (CSR), National Institutes of Health, Oct 2024
9. Expert International Panelist, Croatian Science Foundation, July 2024
10. NSF Panelist, Partnerships for Innovation (PFI): Data Science and Cybersecurity, March 2024
11. NSF Panelist, Small Business Innovation Research (SBIR), Partnerships for Innovation (PFI), Nov 2023
12. NSF Ad-hoc reviewer, Small Business Innovation Research (SBIR), Partnerships for Innovation (PFI), Nov 2023
13. NIH study section reviewer, NIH Director's New Innovator Award Program (DP2), Nov 2023
14. NSF Panelist, Partnerships for Innovation (PFI) program, Oct 2023
15. NIH Study Section Panelist, Clinical Data Management and Analysis Study Section (CDMA), Center for Scientific Review (CSR), National Institutes of Health, Oct 2023
16. NIH Study Section Panelist, Special Emphasis Panel (SEP) for Centers of Biomedical Re- search Excellence (COBRE) Phase 1 (P20), July 2023
17. NSF Panelist, Small Business Innovation Research (SBIR), Partnerships for Innovation (PFI), April 2023
18. NSF Panelist, Cyberinfrastructure for Sustained Scientific Innovation (CSSI), Office of Advanced Cyberinfrastructure (OAC), March 2023
19. Expert Panelist, New Jersey Governor's Council for Medical Research and Treatment of Autism (Council), May 2023
20. NIH Study Section Panelist, Child Psychopathology and Developmental Disabilities (CPDD) study section, Center for Scientific Review (CSR), National Institutes of Health, February 2023
21. NSF Panelist, Partnerships for Innovation (PFI) program, Oct 2022
22. Expert Panelist, New-Jersey (NJ) Governor's Council for Medical Research and Treatment of Autism (Council), Scientific Peer Advisory and Review Services division of the American Institute of Biological Sciences (AIBS), May 2022
23. NIH Study Section Panelist, Clinical Data Management and Analysis Study Section (CDMA), Center for Scientific Review (CSR), National Institutes of Health, June 2022
24. Expert International Reviewer and Panelist, Wellcome Trust, United Kingdom (UK), March 2022
25. NIH Study Section Panelist, NCI Translational and Basic Research Early Lesions (U54 and U24) study section, Center for Scientific Review (CSR), National Institutes of Health, March 2022
26. Expert Reviewer for Natural Sciences and Engineering Research Council (NESRC) of Canada, Jan 2022
27. Expert International Reviewer and Panelist, Belgium Fund for Scientific Research (F.R.S.–FNRS), Jan 2022
28. External Expert for Tenure, Oakland University MI USA Nov 2021
29. Expert Reviewer for Natural Sciences and Engineering Research Council (NESRC) of Canada, Nov 2021
30. NIH Study Section Panelist, NCI Clinical Proteomic Tumor Analysis Consortium (CPTAC) study section, Center for Scientific Review (CSR), National Institutes of Health, Nov 2021
31. External Reviewer for Faculty Tenure, Department of Biology, Lahore University of Management Sciences (LUMS), Lahore Pakistan, Sept 2021
32. Expert Reviewer for NIH Bridge to Artificial Intelligence (Bridge2AI) study section, NIH Com- mon

Fund, Oct 2021
33. NIH Study Section Panelist, Biodata Management and Analysis (BDMA) study section, Cen- ter for Scientific Review (CSR), National Institutes of Health, Oct 2021
34. Expert International Reviewer and Panelist, Belgium Fund for Scientific Research (F.R.S.–FNRS), August 2021
35. NIH Study Section Panelist, National Institute of Neurological Diseases and Stroke NST-2 study panel, Center for Scientific Review (CSR), National Institutes of Health, June 2021
36. NIH Study Section Panelist, Biodata Management and Analysis (BDMA) study section, Cen- ter for Scientific Review (CSR), National Institutes of Health, June 2021
37. NSF Panelist, Small Business Innovation Research (SBIR)/Small Business Technology Trans- fer (STTR) program, Feb 2021
38. Academic Advisor for NIH grants, for Hadi Amini (SCIS FIU), and Shabnam Rezapour Behnagh (FIU), National Research Mentoring Network, University of Utah, Oct 2020 -Jan 2021
39. Expert Reviewer, Oak Ridge National Lab (ORAU), NSF MRI UTSA Limited Submission Program, Oct 2020
40. Expert Reviewer, Oak Ridge Associated Universities (ORAU), University of Texas at San Antonio's (UTSA) Limited Submission Program, June 2020
41. NSF Reviewer, Small Business Innovation Research (SBIR)/Small Business Technology Transfer (STTR) program, May 2020
42. DOE Reviewer, Minority Serving Institution Partnership Program, National Nuclear Security Administration (NNSA), April 2020
43. Expert International Reviewer and Panelist, Belgium Fund for Scientific Research (F.R.S.–FNRS), March 2020
44. Expert Reviewer for Natural Sciences and Engineering Research Council (NESRC) of Canada, Jan 2020
45. Expert Reviewer for NIH NIDDK Chronic Renal Insufficiency Cohort Panel, Scientific and Data Coordinating Center (SDCC) at The University of Pennsylvania, August 2019
46. Expert International Reviewer and Panelist, The Croatian Science Foundation (HRZZ), July 2019
47. NSF Reviewer, Cyberinfrastructure for Sustained Scientific Innovation (CSSI), May 2019 (declined due to COI)
48. External Reviewer, The University of Queensland Diamantina Institute, Feb 2019
49. Reviewer for Croatian Science Foundation (CSF), March 2018
50. NSF Panel, Office of Advanced Cyberinfrastructure (OAC), Oct 2017
51. ORAU NSF Big Data Regional Innovation Hub Panel, 2017
52. Early Career Reviewer (ECR), Center for Scientific Review (CSR), National Institutes of Health (NIH), March 2017 to date
53. Reviewer for Croatian Science Foundation (CSF), March 2014

8.7. Consulting Experience in Accreditation, Review, and Educational Reform

1. Member, Program Accreditation Panels, The National Commission for Academic Accreditation and Assessment (NCAAA) for Computer Science Department, King Faisal University, Saudi Arabia

8.8. University and Department service

8.8.1. Florida International University (FIU)

Mentor and evaluation for Junior Faculty

- Mentor for Olaoluwa Adigun, KFSCIS FIU, Sept 2025 - to date
- Peer-teaching evaluation of Farhad Shirani (COT-3100 Discrete Structures), Fall 2023.
- Peer-teaching evaluation of Agoritsa Polyzou (CAP 5771: Principles of Data Mining), Fall 2022.
- Mentor for Hadi Amini, SCIS FIU, Nov 2020 - May 2025 (Dr. Amini got his tenure in May 2025)
- Member of FIU Faculty Mentor Program, (Mentor for Dr. Imran Ahmed, Research Assistant Professor | Food, Agriculture, and Bio. Innovation Lab (FABIL), Sept 2020 -
- Member of FIU Faculty Mentor Program, (Mentor for Dr. Xuan Lv, Assistant Professor, CONST, EC), Sept 2018 -

Panelist and Judging Service

- Panelist for SCIS retreat (Research & Funding), Jan 2023

- Panelist for SCIS Professional Development Workshop Series for Junior Faculty, Sept 2020
- Judging in the BSI symposium, Department of Physics FIU, July 2020
- Judge for SCIS Research Day, SCIS FIU, Oct 2019

Committee Service

- KFSCIS Tenure and Tenure-Track Faculty Search and Screen Committee (TTT Hiring), FIU, August 2025 - May 2026
- Chair, KFSCIS Infrastructure Committee, FIU, August 2025 - May 2026
- AI Research Subcommittee, FIU Experience Impact 2030 Strategic Plan Implementation, 2025
- Chair, Ad hoc committee for determining Graduate Faculty (GF) status, KFSCIS FIU, 2025
- **Chair**, *Research Commercialization and Industry Partnerships sub-committee*, **Provost's Strategic 2030 Plan Implementation Committee**, 2025
- Member of **Provost's Strategic Plan (2025-2030) Implementation Committee (Research Excellence Pillar)**, 2025
- KFSCIS CEC building ad-hoc committee, Oct 2024
- CEC Faculty Council Representatives, FIU, August 2024 - May 2025
- CEC T&P Committee Representatives, FIU, August 2024 - May 2025
- KFSCIS Tenure and Tenure-Track Faculty Search and Screen Committee (TTT Hiring), FIU, August 2024 - May 2025
- Chair, KFSCIS Infrastructure Committee, FIU, August 2024 - May 2025
- KFSCIS Representative to the Faculty Council, FIU, May 2024 - May 2026 (KFSCIS level elected)
- Foreign Language Task Force, Faculty Senate FIU, Feb 2024 - to date (University level Elected)
- Chair KFSCIS Infrastructure Committee, FIU, June 2023 - June 2024
- Chair of Diversity and Inclusion Committee, School of Computing, FIU, June 2022 - June 2023
- Member of Director Search, Knight Foundation School of Computing and Information Sciences (KFSCIS), 2023
- Member of Tenure Track Faculty Hiring Committee, Knight Foundation School of Computing and Information Sciences (KFSCIS) at FIU (2020, 2021, 2022)
- CEC Tenure and Promotion (T&P) Committee (College level Elected), May 2022 - May 2024
- ABET evaluation and assessment of CDA 3102 (Computer Architecture) course, Dec 2021
- CEC Faculty Senate and University Committee Representative, DAS Evaluation Committee Representative (Elected), May 2021
- Member of Tenure Track Faculty Hiring Committee, Department of Biomedical Engineering at FIU, Dec 2020
- Member of Diversity and Inclusion Committee, School of Computing, FIU, Sept 2020 - May 2022
- Led a team of 5 FIU researchers for Faculty Grantsmanship Development Program –2019- 2020. Allowed junior researchers to attend NIH/NSF grant workshops and a seed grant from the FIU Research & Economic Development office.
- Member of Eminent Scholar Chair Promotion Committee, SCIS FIU, August 2019
- Member of College of Engineering and Computing (CEC) Scholarship Committee, April 2019
- Gave a talk entitled "Interdisciplinary Big Data Computational Sciences and Opportunities", at the School of Computing and Information Sciences (SCIS) Graduate Program Open House & Orientation, Aug 2018
- Member of SCIS Graduate Committee, Aug 2018 - August 2020
- Member of SCIS Tenure Track Faculty Hiring Committee, Aug 2018 - August 2023
- Member of SCIS ACM Student Chapter Faculty Advisor, Aug 2018 - to date

8.8.2. Western Michigan University (WMU)

- Panelist for NSF CAREER Workshop for Faculty, Office of Vice President of Research (OVPR), WMU, March 2018
- Member of CEAS Faculty Creative Initiative Review Committee, WMU, Feb 2018
- Member of Undergraduate Computer Engineering Curriculum Revision Committee, ECE Department, Sept 2017 - to date
- Panelist for Discover Discovery Workshop for New Faculty, Office of Vice President of Re- search (OVPR), WMU, Sept 2017
- Member of committee on designing and building Biomedical Engineering Program for CEAS

WMU, July 2017 - to date
- Member of CEAS outstanding awards nomination, Jan 2017
- Member of Graduate Committee, CS Department, Sept 2016 - to date
- Member of Undergraduate Computer Engineering Curriculum Revision Committee, ECE Department, Sept 2016 - Dec 2016
- Member sub-committee for research Strategic Plan for College of Engineering and Applied Science (CEAS) WMU, March 2016
- Member of Strategic Plan Committee for College of Engineering and Applied Science (CEAS) WMU, March 2016
- Panelist for Discover Discovery Education and Training for New Faculty, Office of Vice Pres- ident of Research (OVPR), WMU, Sept 2015
- Member of Undergraduate Computer Engineering Curriculum Revision Committee, ECE Department, Oct 2014 - Aug 2015
- Member of Seminar Committee, ECE Department, Oct 2014 - to date
- Member of Computing Hardware Committee, Computer Science Department, Sept 2014 - to date
- Proctoring service for ECE Qualifying Exam, March 2014
- ECE representative for the College of Engineering and Applied Sciences (CEAS) Scholarship Committee, Feb 2014 - to date

8.8.3. Outreach and Community Service (recent)

- International expert on Curriculum Revision, University of Engineering & Technology (UET) Lahore June 2020 - June 2021 (1 meeting for per month among faculty at UET Lahore, Iowa State University, Michigan University, and Industry including Intel, Argus Network)
- Participant for NSF Diversity Advocate group at FIU, Dec 2019
- Participant for FIU Bystander Leadership Workshop, Sept 2019
- Judge for Regional Intel Science and Engineering Fair, for high school students (K-12) Kalamazoo Area Math and Science Center (KAMSC), March 2017
- Judge for 16th Annual Southwest Michigan Regional (Intel Science and Engineering) ISEF Science Fair for high school students (K-12) Kalamazoo Area Math and Science Center (KAMSC), March 2016
- Judge for Southwest Michigan Science and Engineering Fair for high school students (K-12) Kalamazoo Area Math and Science Center (KAMSC), March 2015
- Judge for Regional Science and Engineering Fair for high school students (K-12) Kalamazoo Area Math and Science Center (KAMSC), March 2014

8.8.4. Miscellaneous

- Judge in genomics study section for Fellows Award for Research Excellence (FARE), National Institutes of Health (NIH) 2013-2014
- Student representative for Electrical and Computer Engineering Department, Graduate Stu- dent Council, University of Illinois at Chicago (Jan 2008 to Summers 2010).

9. Consultation Experience

- **Dialectica** Athens Greece

Medical devices department Feb 2020

Consulted for London UK based company who wanted to enter the clinical space related to novel molecular, therapeutics and computational techniques for US hospitals and clinics.

- **Western Michigan University** Kalamazoo, MI USA

Computer Science Department June 2019 to July 2019

Provided expert consultation on developing machine-learning models for high-dimensional mass spectrometry data. Also advised on high-performance computing solutions to these problems and on systems biology aspects of the work.

- **Chulalongkorn University** Bangkok, Thailand

Systems Biology (CUSB) Center Oct 2013 to Nov 2013

This work involved design of setting up a compute infrastructure for large-scale system biology computational needs. A memory-distributed compute cluster, a high-performance work-station, an intelligent memory system and GPU’s were part of the setup.

- **University of Illinois at Chicago (UIC)** Chicago, IL USA

Parallel algorithms and multimedia systems laboratory May 2010 to August 2010
This work involved design of setting up a high-performance compute clusters for computational biology algorithms. A 16-node system was setup as a pilot project.

10. Patents

1. Fahad Saeed, and Umair Mohammad “Systems and Methods for Patient-specific Epileptic Seizure Prediction”, United States Patent and Trademark Office (USPTO), U.S. Department of Commerce, Feb 10th 2026 (***US Patent # US 12,544,002 B2***)
2. **Fahad Saeed**, and Sumesh Kumar, “Systems and Methods for matching mass spectrometry data with a peptide database”, United States Patent and Trademark Office (USPTO), U.S. Department of Commerce, Dec 12, 2023 (***US Patent # US 11,842,799 B1***)
3. **Fahad Saeed**, and Sumesh Kumar, “Systems and Methods for matching mass spectrometry data with a peptide database”, United States Patent and Trademark Office (USPTO), U.S. Department of Commerce, Oct 24, 2023 (***US Patent # US 11,798,654 B1***)
4. **Fahad Saeed**, and Fahad Almuqhim, “Systems and Methods for Diagnosing Autism Spec- trum Disorder Using FMRI Data”, United States Patent and Trademark Office (USPTO), U.S. Department of Commerce, July 5th, 2022 (***US Patent # 11,379,981 B1***)
5. **Fahad Saeed**, and Muhammad Haseeb, “Systems and methods for peptide identification”, United States Patent and Trademark Office (USPTO), U.S. Department of Commerce, April 19th, 2022 (***US Patent # 11,309,061 B1***)
6. **Fahad Saeed**, and Muhammad Usman Tariq, “Systems and methods for measuring similarity between mass spectra and peptides”, United States Patent and Trademark Office (USPTO), U.S. Department of Commerce, February 15, 2022 (***US Patent # 11,251,031***)
7. **Fahad Saeed**, and Muhammad Haseeb, “Methods and Systems for Compressing Data”, United States Patent and Trademark Office (USPTO), U.S. Department of Commerce, October 20, 2020 (***US Patent # 10,810,180***)

10.1. Provisional Patent Applications

- “SPERTL: Epileptic Seizure Prediction using EEG with ResNets and Transfer Learning”, FIU Technology Management and Commercialization, USPTO # 18/341,967, Filed June 27, 2023

10.2. Invention Disclosures & Non-Provisional Patent Applications

- “Predicting peptide properties from mass spectrometry data using deep attention-based multitask network and uncertainty quantification”, FIU Technology Management and Commercialization, Reference No. Disclosure D2024-0064
- “GPU based database search for peptide identification from large-scale mass spectrometry data”, FIU Technology Management and Commercialization, Reference No. Disclosure D2023-0041
- “SPERTL: Epileptic Seizure Prediction using EEG with ResNets and Transfer Learning”, FIU Technology Management and Commercialization, Reference No. Disclosure D2022-0051
- “Systems and Methods for matching mass spectrometry data with a peptide database”, FIU Technology Management and Commercialization, Reference No. Disclosure 2021-077
- “ASD-SAENet: Sparse Autoencoder for detecting autism spectrum disorder (ASD) using fMRI data”, FIU Technology Management and Commercialization, Reference No. Disclosure D2021-0015
- “SpeCollate: Deep Cross-Modal Similarity Network for Mass Spectrometry Data based Pep- tide Deductions”, FIU Technology Management and Commercialization, Reference No. Dis- closure D2021-0010
- “HPC methods for peptide deduction on distributed memory architectures”, FIU Technology Management and Commercialization, Reference No. Disclosure D2020-0114, Oct 2020
- “Efficient Shared Peak Counting in Database Peptide Search Using Compact Data Structure for Fragment-Ion Index”, FIU Technology Management and Commercialization, Reference No. Disclosure D2019-0056, Oct 2019
- “A Deep-Learning and fMRI data-based approach for diagnosis of Autism Spectrum Dis- order”, FIU Technology Management and Commercialization, Reference No. Disclosure D2019-0044, Oct 2019
- “Dynamic programming algorithm for phosphorylation site assignment”, Office of Technology Transfer

and Development at National Institutes of Health (NIH)

- “Multiple sequence alignment system for pyrosequencing reads”, Office of Technology Management at University of Illinois at Chicago, Reference No. DC091,
- “Sample-Align-D: A High-Performance Multiple Sequence Alignment System using Phyloge- netic Sampling and Domain Decomposition”, Office of Technology Management at University of Illinois at Chicago, Reference No. DC082

11. Continuing Professional Education

- NSF Regional ICorps, NYC Cohort, Spring 2023
- NSF National iCorps, NYC Cohort, Fall 2022
- NSF Regional iCorps, UC Berkeley Cohort, Summer 2022
- Bystander Leadership workshop, FIU, 2019
- FIU Diversity Advocate Workshop, 2021
- STRIDE Training (Diversity and Inclusion) Workshop, 2021

12. Professional Membership

- Institute of Electrical and Electronics Engineers (IEEE), since 2005
- Association for Computing Machinery (ACM), since 2010